Exploring Social Interest in Burnout Using Topological Data Analysis on Google Trends Data

Keywords: *burnout, job stress, emotional exhaustion, depression, autistic burnout, work burnout, Topological Data Analysis, Google Trends*

1. Introduction

Burnout is a global phenomenon that affects a wide range of individuals, including teachers, nurses, students, and probation officers (Avola et al., 2025; Diaz et al., 2025; Munn et al., 2025; Galanis et al., 2025; Abraham et al., 2024; Alward & Viglione, 2025). Mainly linked with occupational consequences (Shanafelt et al. 2009; Borritz et al. 2006), it has adverse physiological effects along with its common mental health symptoms (Honkonen et al. 2000; Khoshakhlagh et al., 2024). While burnout has always caused concerns, its prevalence surged during COVID-19 and continues with its aftermath (Gaspar et al., 2024 & Abraham et al., 2024). With increased interest in mental health, TDA offers a novel perspective for understanding complex temporal data (Carlsson, 2020; Godwin et al., 2019). This study aims to explore the topological structure of social interest in burnout using Topological Data Analysis (TDA) on Google Trends (GT) data in the United Kingdom.

2. Literature Review

2.1. Topological Data Analysis

Backed by computational tools and mathematical techniques, TDA is a growing field of data analysis (Chazal & Michel, 2021). It investigates topological and geometric methods to extract reliable insights from “noisy dynamical structures” (Rudkin et al., 2023). Its applications range from chemistry (Smith et al., 2021), material science (Pike et al., 2020), and medicine (Dindin et al., 2020) to multivariate time series analysis (Khasawneh and Munch, 2016; Seversky et al., 2016; Umeda, 2017), sensor networks (De Silva and Ghrist, 2007), image analysis (Qaiser et al., 2019; Rieck et al., 2020), 3D shape analysis (Skraba et al., 2010; Turner et al., 2014), transportation (Li et al., 2019), and finance (Gidea et al., 2020). Recently, the application of TDA has grown in investment (Goel et al., 2020), trade

(Guo et al., 2022), stock markets (Yen and Cheong, 2021), and forecasting market trends (Gidea and Katz, 2018; Rudkin et al., 2023).

2.2. Burnout and Google Trends

Popular searches on the internet are a reflection of public interest (Wilson, 2018). Currently, big data offers valuable information to understand and create dependable proxies for social indicators (Di Bella et al. 2018). By using "burnout" and related search terms, we aim to evaluate social interest over the past five years in the UK. The most effective way to identify related search terms is through GT's related queries feature. However, the top related searches—such as "the burnout," and "burnout paradise"—offer limited relevance. While these queries reflect public interest, utilizing them will not give us a comprehensive understanding of the area of burnout we are focusing on (i.e., occupational burnout and mental health).

Burnout has a positive correlation with job stress (Khalatbari et al., 2013), is characterized as emotional exhaustion (Aguilera et al., 2021) and has a significant overlap with depression (Schonfeld & Bianchi, 2016; Iacovides et al., 2003). Therefore, the first three keywords were exchanged with "job stress", "emotional exhaustion", and "depression".

3. Methodology

TDA is increasingly applied to time series data (Ravishanker & Chen, 2019). Two common topological techniques are: the persistence diagram, which tracks loops and holes across varying scales, and the mapper graph, which captures the data's overall shape in a 1D structure useful for exploration and visualization (Munch, 2017). Other features like persistent homology identifies topological features by marking their births and deaths over changing distances (Carlsson, 2009), while persistence norms reduce this information to numerical values that reflect the data's structural complexity (Dłotko & Rudkin, 2023).

According to Dłotko & Rudkin (2023), data is represented as a point cloud where each point captures multiple variables. As the scale increases, topological features like clusters (dim 0) and loops (dim 1) emerge and fade. The norms for dimension g ∈ {0, 1} are calculated using:

$$L_{g1} = \sum_{f=1}^{F_g} d_f - b_f$$

and:

$$L_{g2} = \sqrt{\sum_{f=1}^{F_g} (d_f - b_f)^2}$$

Where F_g represents the number of features in homology dimension g, and birth (b_f) and death (d_f) filtration values can show a feature's lifetime.

To track changes in the data's topological structure over time, Wasserstein distance is used (Ravishanker & Chen, 2019). This metric measures how much effort is needed to match features between diagrams, creating a time series that reflects week-by-week topological shifts. It is defined as:

$$\mathbf{W}_{q,\tilde{p}}(\tilde{\Omega}_1, \tilde{\Omega}_2) = \Big[\inf_{\eta:\tilde{\Omega}_1 \to \tilde{\Omega}_2} \sum_{\tilde{\tau}_{\tilde{p},k} \in \tilde{\Omega}_1} |\tilde{\tau}_{\tilde{p},k} - \eta(\tilde{\tau}_{\tilde{p},k})|_\infty^q \Big]^{1/q}, q = 1, 2, \ldots,$$

Where p is dimension and q is degree. (Ravishanker & Chen, 2019).

4. Data

4.1. Description of the Data

Weekly GT data for the past five years (April 2020 to 2025) was used for six keywords, leading to 261 entries per keyword (Table 1).

Table 1

Keywords and Interest by Subregion

		Interest by Subregion			
S.No.	Keyword	England	Scotland	Wales	Northern Ireland
1.	burnout	100	96	93	80
2.	job stress	86	80	93	100
3.	emotional exhaustion	66	66	66	100
4.	depression	85	84	100	98
5.	autistic burnout	81	81	100	54
6.	burnout work	100	90	90	80

Note: The score shows the search terms relative popularity during the last five years. It ranges from 0 to 100, with 100 representing the location where the term had the highest share of total searches. A score of 50 means half that popularity, while 0 means there wasn't enough data. Higher scores reflect a greater proportion of local searches, not the total number.

Even though Scotland has the second highest population and land area (Neil, 2022), there are no keywords that scored a 100. Northern Ireland being the smallest in both aspects, scored a 100 in job stress and emotional exhaustion. Thus, Northern Ireland shows more focused relative interest whereas Scotland's search behavior is more diluted across other topics.

4.2. Patterns of Public Interest in Burnout in the UK

Table 2

Summary Statistics

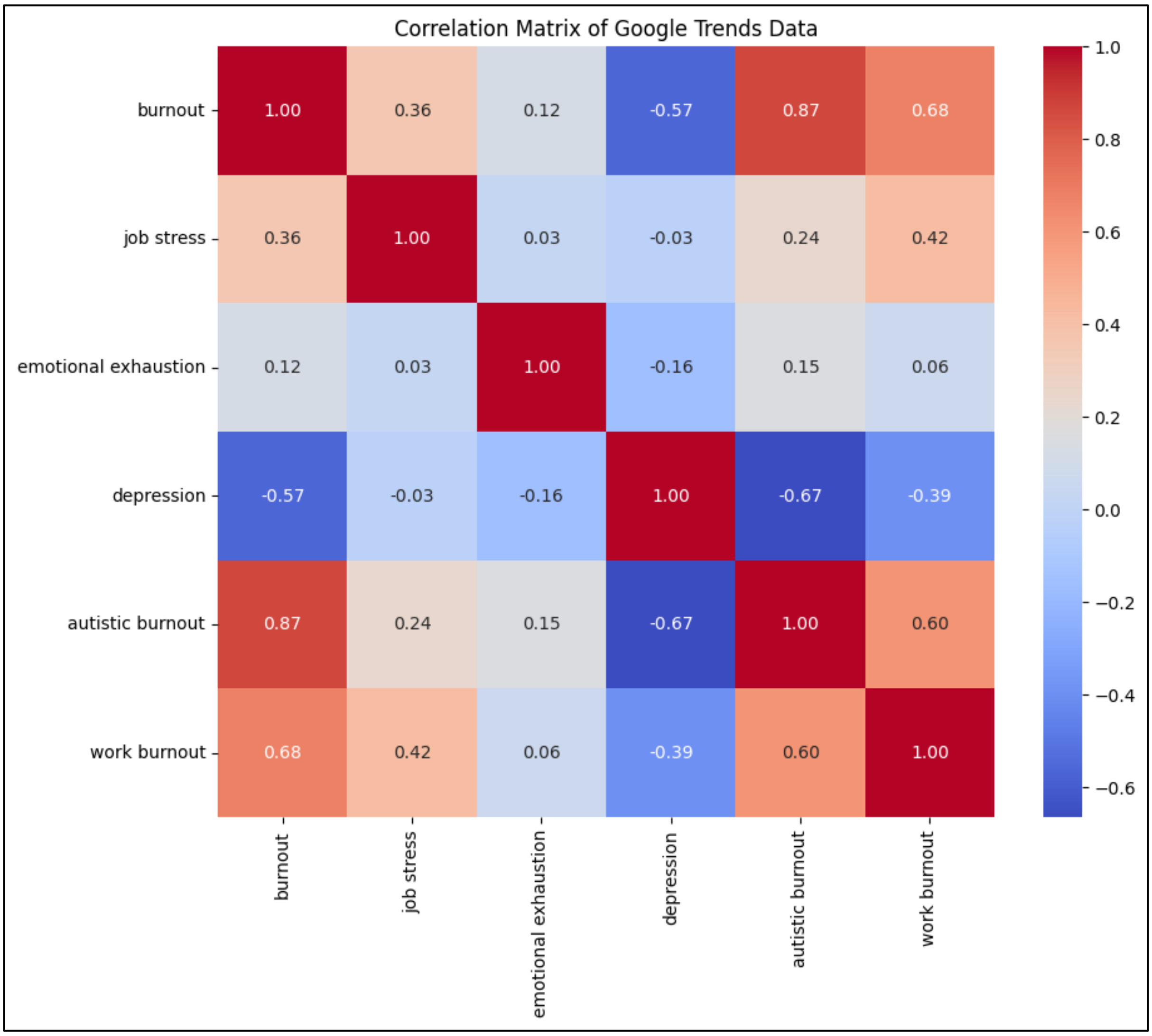
Search Terms	Mean	Min	Q25	Q50	Q75	Max	SD	Skew	Kurt
burnout	58.88	31	47	57	71	100	14.55	0.33	-0.76
job stress	54.84	0	45	54	66	100	16.19	-0.52	1.64
emotional exhaustion	3.53	0	0	0	0	100	12.33	3.96	18.36
depression	73.41	50	64	71	83	100	11.63	0.24	-0.96
autistic burnout	31.78	0	0	33	56	100	28.86	0.18	-1.37
burnout work	45.69	0	33	50	67	100	26.62	-0.51	-0.70

Note: Descriptive statistics for six search terms linked to burnout, based on Google Trends data collected in the United Kingdom from April 2020 to April 2025. The values range from 0 to 100, reflecting the Relative Search Index (RSI) computed by Google. Abbreviations used: Min for minimum value, Q25 for 25th percentile, Q50 for median, Q75 for 75th percentile, Max for maximum value, SD for standard seviation Skew as a measure of the skewness and Kurt as the kurtosis.

Through Table 2, we understand the spread of interest amongst these terms. For instance, the *minimum* value for depression is 50, showing extremely high interest in this topic. Mean values also show high interest in depression, followed by burnout, job stress and work burnout respectively.

Figure 1

Correlation Matrix of Google Trends Data

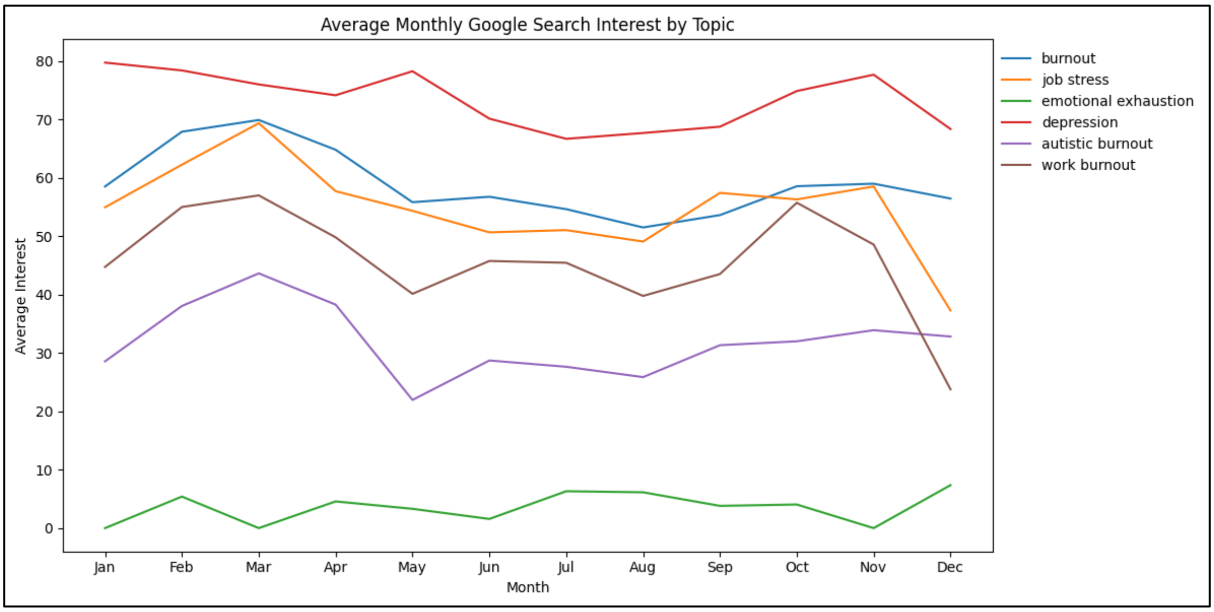


Note: Figure shows a correlation matrix as a heatmap.

Figure 1 shows high correlations in autistic burnout, work burnout, and burnout, and a negative correlation with depression.

Figure 2

Average Monthly Searches

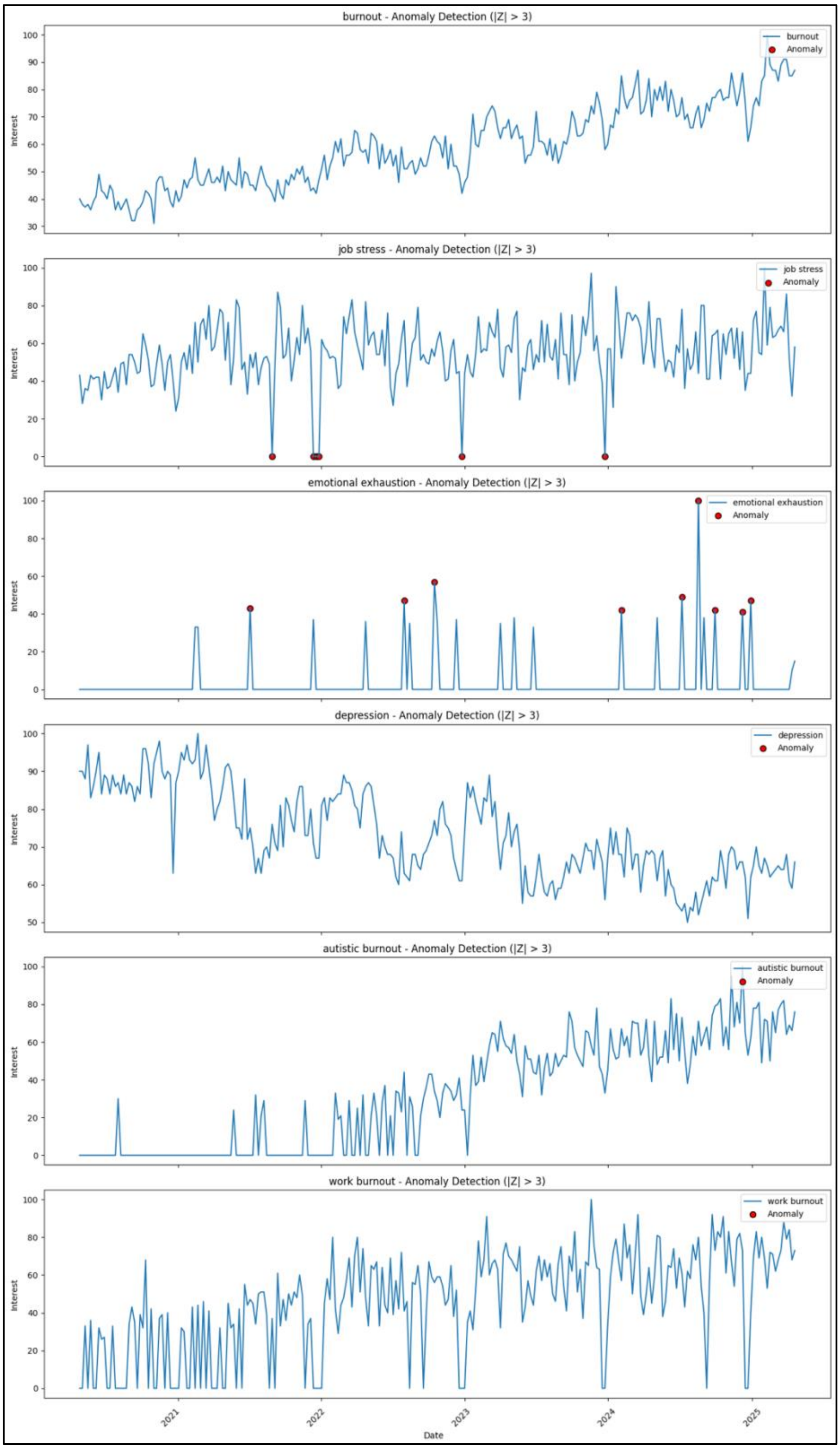


Note: Figure shows average monthly google searches for all six search terms.

In Figure 2, it is observed that burnout, job stress, work burnout, and autistic burnout peaked during March. The popularity of the first three terms could be attributed to the fiscal year ending in March/early April (Wikipedia, 2025). This might lead to high occupational stress leading to a higher search interest.

Figure 3

Anomaly Detection in Time Series Data



Note: Figure shows anomalies as red dots (only observed in job stress and emotional exhaustion)

Anomalies can often indicate important moments in time (Schmidl et al., 2022). Figure 3 shows that job stress has anomalies with small values, whereas emotional exhaustion has anomalies as high values which may be a reflection of the interest in these terms.

4.3. TDA (Time Series and Point Clouds A and C)

Figure 4

Time series for Burnout and Mental health terms

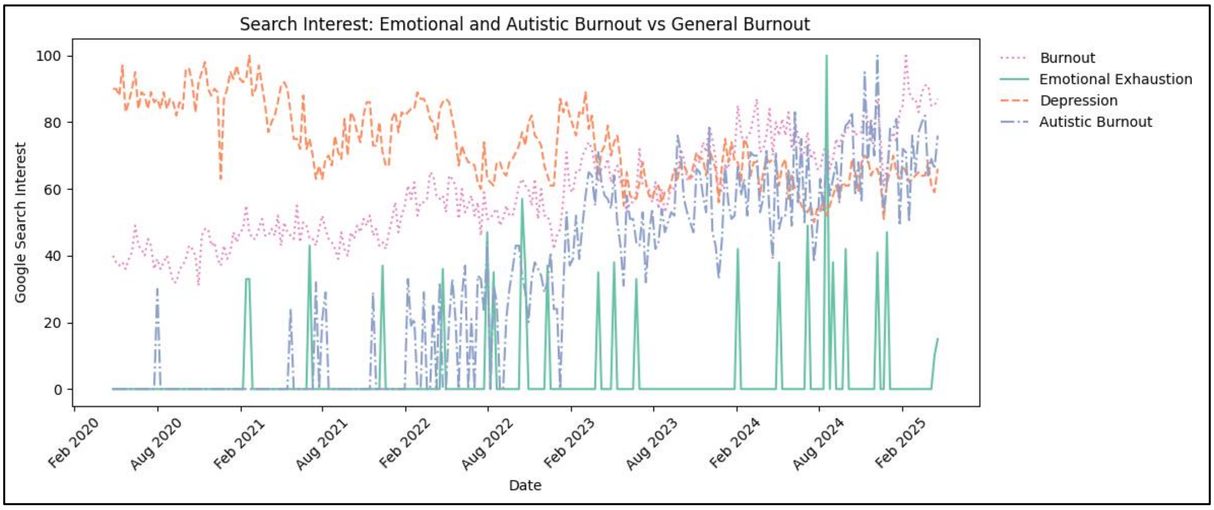
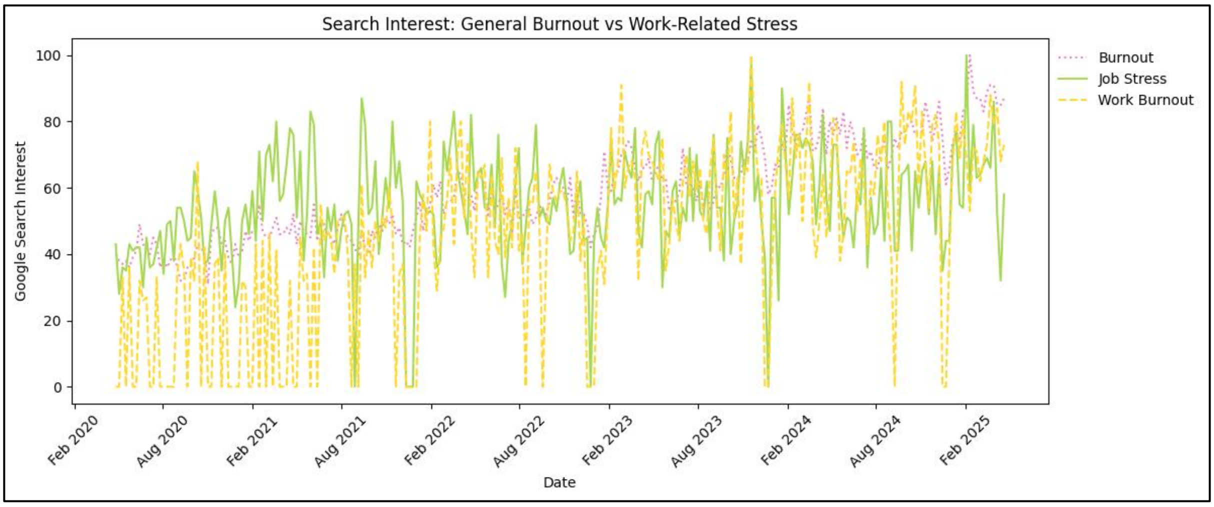


Figure 5

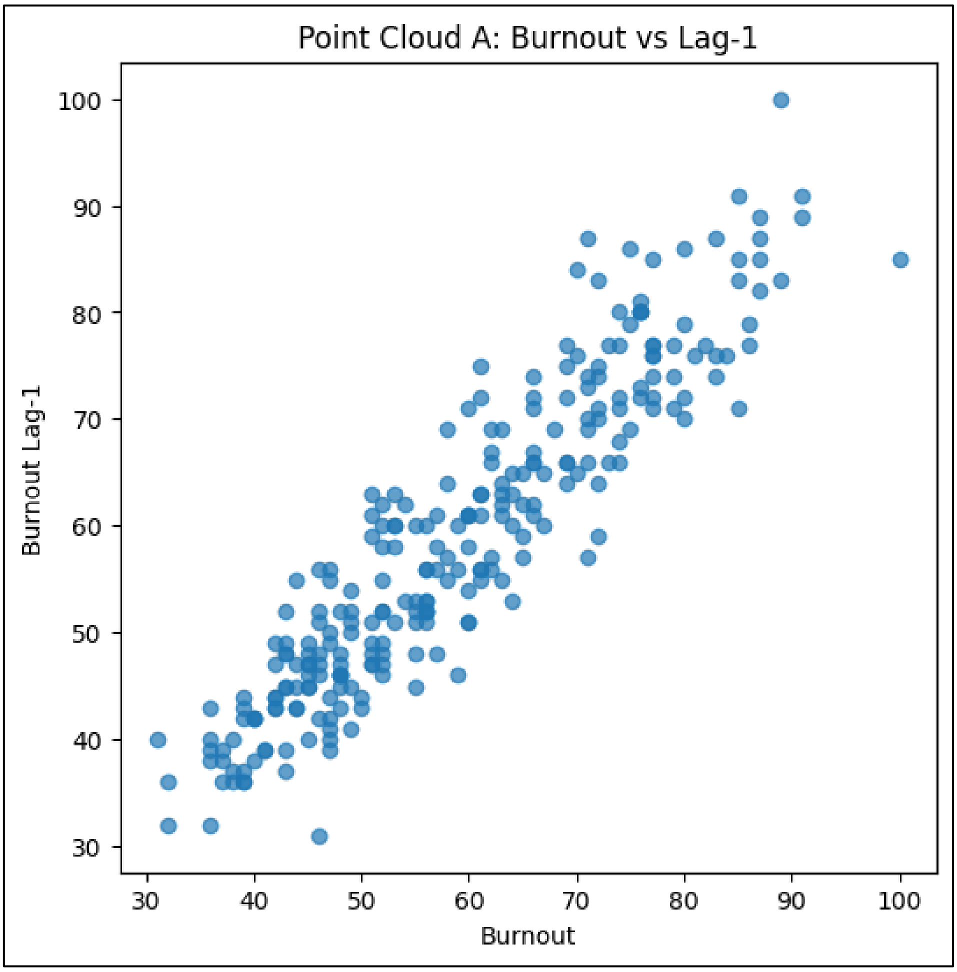
Time series for Burnout and Occupation related terms



Figures 4 and 5 show time series plots with burnout and different focus areas (mental health and occupation), over six-month intervals. Autistic burnout saw high interest in 2023 and has since matched depression and burnout in search frequency (Figure 4). Job stress and work burnout seem to rise and fall together (Figure 5). Looking at both graphs, occupation related search terms perform better with burnout.

Figure 6

Point Cloud A: Keyword plus 1 lag



Note: Figure shows internal consistency of search terms through a one-week lag.

Figure 7

Keyword plus 4 lags

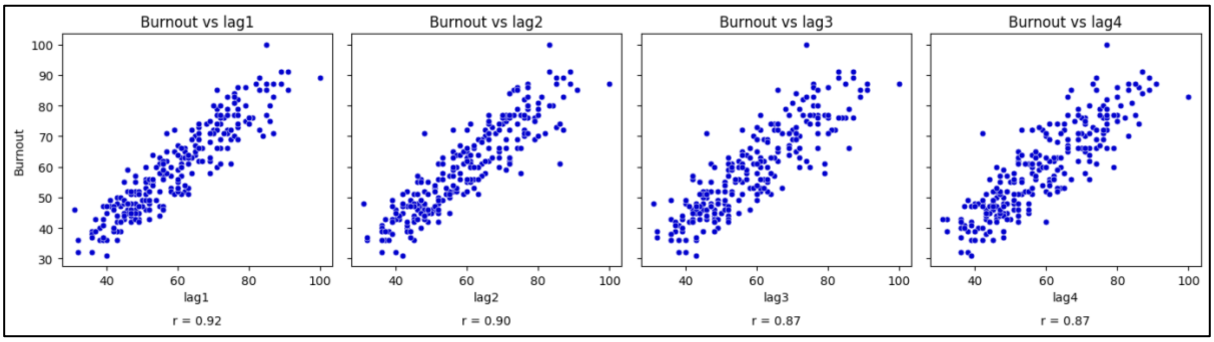
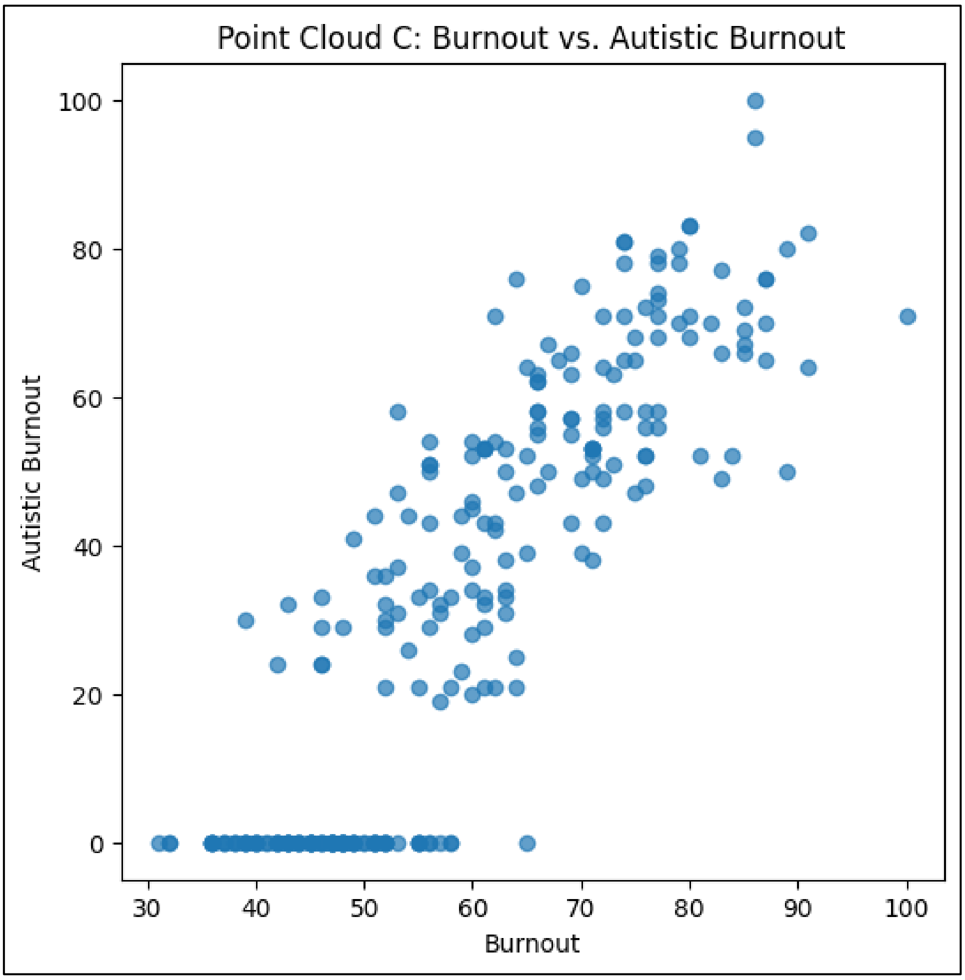


Figure 8

Point Cloud C: Keyword plus 1 related search



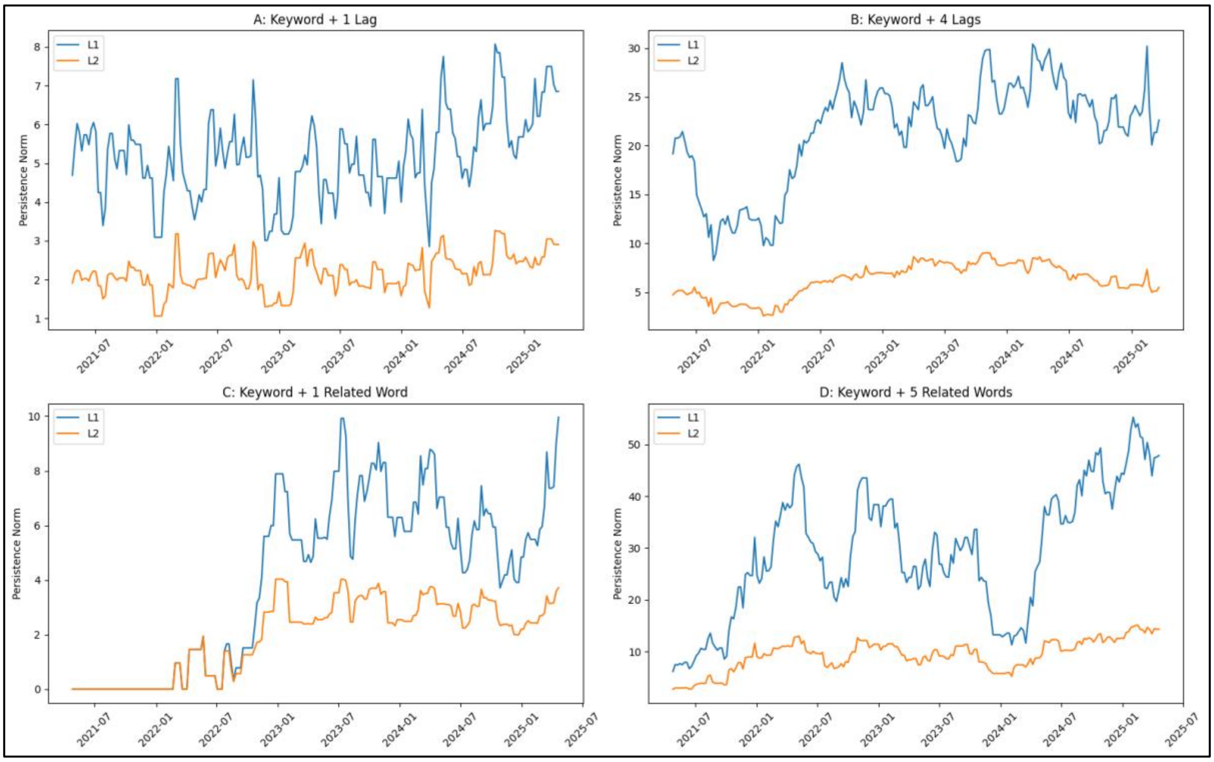
Note: Figure shows point cloud for keyword (burnout) and 1 related search (autistic burnout). This search term was chosen due to its high correlation with burnout, 0.87)

The diagonal line in Figure 6 and 7 suggests consistent search patterns weekly (See Appendix for persistence diagram). The collection of points in Figure 8 reflects the closeness of search term and related term showing similarity in the perception of both terms amongst UK citizens.

5. Results

Figure 9

Time series of persistence norms for 52-week window

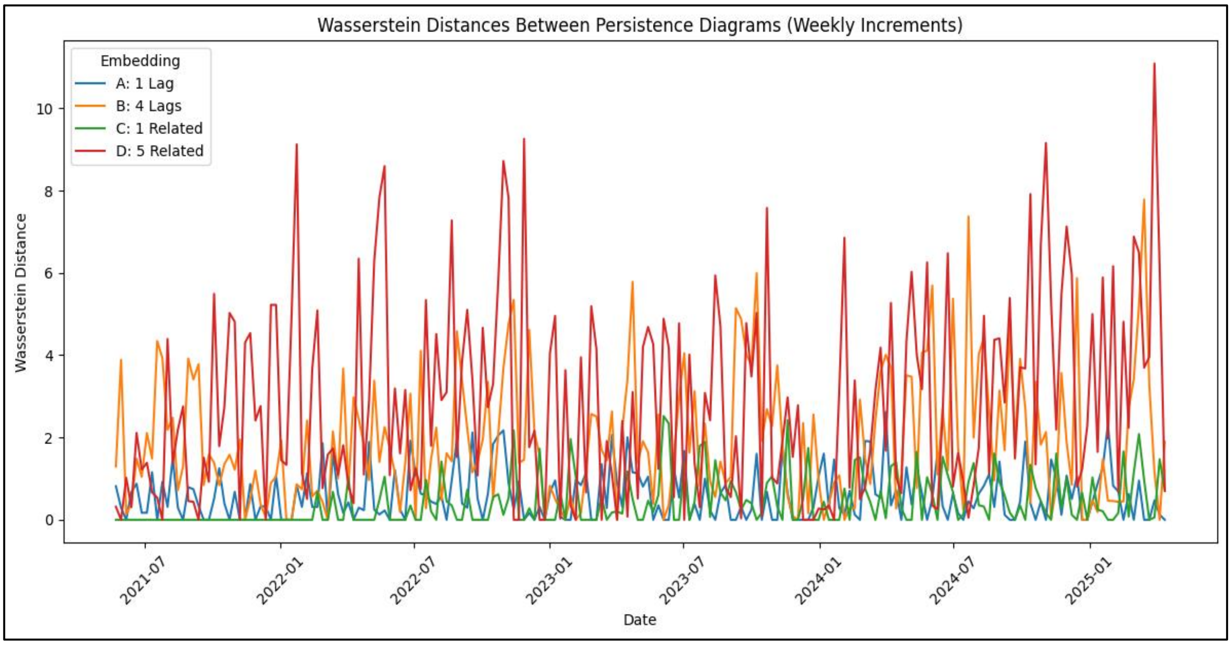


Note: Figure shows Time series of persistence norms for 52-week window with- A: Keyword plus 1 lag, B: Keyword plus 4 lags, C: Keyword plus 1 related search, D: Keyword plus 5 related searches

Figure 9 shows the evolution of the topological complexity over time with different embeddings. Plots A and B show relatively stable and lower norm values, resulting in more simpler and consistent topological structures. Plots C and D show a contrasting image with a sharp rise in both L1 and L2 norms. Plot D especially has high norms hinting that the structural complexity increases as more dimensions (i.e., related terms) are added. Greater embedding complexity (from A to D) correlates with higher persistence norms, especially in dimension 1, suggesting more persistent and complex topological features.

Figure 10

Time series of Wasserstein Distances between Persistence Diagrams



Note: Figure shows time series of Wasserstein distances between persistence diagrams with 1-week increments for A to D

Figure 10 captures the degree of change in the topological structure weekly. Embedding D shows the highest and most volatile distances, suggesting rapid shifts in the underlying data patterns when multiple terms are considered.

6. Conclusion

Burnout is a complex phenomenon. We wanted to understand whether the public views burnout as more of a mental health concern or work-related concern. The consistency in burnout over lag clouds and similar search patterns of work-related terms like job stress and work burnout all point towards the fact that people perceive burnout as an occupational event. This aligns well with the current literature as the WHO classified burnout as an occupational phenomenon in the 11th edition of the International Classification of Diseases (ICD-11) (Edú-Valsania et al., 2022). TDA has proved to be an excellent tool to understand complex time series data and further research can involve applying more statistical tests and exploring other homology dimensions to get more intricate structures.

Word Count: 1087

References

Abraham, A., Chaabna, K., Sheikh, J. I., Mamtani, R., Jithesh, A., Khawaja, S., & Cheema, S. (2024). Burnout increased among university students during the COVID-19 pandemic: A systematic review and meta-analysis. Scientific Reports, 14(1), 1-11. https://doi.org/10.1038/s41598-024-52923-6

Alward, L. M., & Viglione, J. (2025). Individual Characteristics and Organizational Attributes: An Assessment of Probation Officer Burnout and Turnover Intent. International Journal of Offender Therapy and Comparative Criminology. https://doi.org/10.1177/0306624X231159882

Aguilera, A. M., Fortuna, F., Escabias, M., and Di Battista, T. (2021). Assessing social interest in burnout using google trends data. Social Indicators Research, 156:587–599.

Avola, P., Soini-Ikonen, T., Jyrkiäinen, A. *et al.* (2025). Interventions to Teacher Well-Being and Burnout A Scoping Review. *Educ Psychol Rev* **37**, 11 https://doi.org/10.1007/s10648-025-09986-2

Borritz, M., Rugulies, R., Christensen, K., Villadsen, E., & Kristensen, T. (2006). Burnout as a predictor of self-reported sickness absence among human service workers: Prospective findings from three year follow up of the PUMA study. Occupational and Environmental Medicine, 63, 98–106.

Carlsson, G. (2009). Topology and data. Bulletin of the American Mathematical Society, 46(2), 255–308. https://doi.org/10.1090/S0273-0979-09-01249-X

Carlsson, G. (2020). Topological methods for data modelling. Nature Reviews Physics, 2(12), 697-708. https://doi.org/10.1038/s42254-020-00249-3

Chazal, F., & Michel, B. (2021). An Introduction to Topological Data Analysis: Fundamental and Practical Aspects for Data Scientists. Frontiers in Artificial Intelligence, 4, 667963. https://doi.org/10.3389/frai.2021.667963

De Silva, V., and Ghrist, R. (2007). Homological Sensor Networks. Notices Am. Math. Soc. 54 (1).

Di Bella, E., Leporatti, L., & Maggino, F. (2018). Big data and social indicators: Actual trends and new perspectives. Social Indicators Research, 135(3), 869–878

Diaz, M., Garcia, Y., Ré, T., & Szabo, T. (2025). Transforming strain into strength: Alleviating stress and burnout in special education teachers with ACT matrix intervention. Journal of Contextual Behavioral Science, 35, 100870. https://doi.org/10.1016/j.jcbs.2025.100870

Dindin, M., Umeda, Y., Chazal, F. (2020). Topological Data Analysis for Arrhythmia Detection Through Modular Neural Networks. In: Goutte, C., Zhu, X. (eds) Advances in Artificial Intelligence. Canadian AI 2020. Lecture Notes in Computer Science (), vol 12109. Springer, Cham. https://doi.org/10.1007/978-3-030-47358-7_17

Dłotko, P., & Rudkin, S. (2023). Persistence norms and the datasaurus. arXiv preprint arXiv:2309.13479. https://arxiv.org/abs/2309.13479

Edú-Valsania, S., Laguía, A., & Moriano, J. A. (2022). Burnout: A Review of Theory and Measurement. International Journal of Environmental Research and Public Health, 19(3), 1780. https://doi.org/10.3390/ijerph19031780

Galanis, P., Moisoglou, I., Katsiroumpa, A., Gallos, P., Kalogeropoulou, M., Meimeti, E., Vraka, I. (2025). Workload increases nurses' quiet quitting, turnover intention, and job burnout: evidence from Greece[J]. AIMS Public Health, 2025, 12(1): 44-55. doi: 10.3934/publichealth.2025004

Gaspar, T., Botelho-Guedes, F., Cerqueira, A. et al. (2024). Burnout as a multidimensional phenomenon: how can workplaces be healthy environments? J Public Health (Berl.) https://doi.org/10.1007/s10389-024-02223-0

Gidea, M., Goldsmith, D., Katz, Y., Roldan, P., & Shmalo, Y. (2020). Topological recognition of critical transitions in time series of cryptocurrencies. Physica A: Statistical Mechanics and its Applications, 548, 123843. https://doi.org/10.1016/j.physa.2019.123843

Godwin, A., & Thielmeyer, A. R. H., & Rohde, J. A., & Verdin, D., & McIntyre, B. B., & Baker, R. A., & Doyle, J. (2019). Using Topological Data Analysis in Social Science Research: Unpacking Decisions and Opportunities for a New Method Paper presented at 2019 ASEE Annual Conference & Exposition , Tampa, Florida. 10.18260/1-2—33522

Goel, A., Pasricha, P., & Mehra, A. (2020). Topological data analysis in investment decisions. Expert Systems With Applications, 147, 113222. https://doi.org/10.1016/j.eswa.2020.113222

Honkonen, T., Ahola, K., Pertovaara, M., Isometsa, E., Kalimo, R., & Nykyri, e a E. (2000). The association between burnout and physical illness in the general population—Results from the finnish health 2000 study. Journal of Psychosomatic Research, 61(59–66), 2006.

Iacovides, A., Fountoulakis, K., Kaprinis, S., & Kaprinis, G. (2003). The relationship between job stress, burnout and clinical depression. Journal of Affective Disorders, 75(3), 209-221. https://doi.org/10.1016/S0165-0327(02)00101-5

Khalatbari, J., Ghorbanshiroudi, S., & Firouzbakhsh, M. (2013). Correlation of Job Stress, Job Satisfaction, Job Motivation and Burnout and Feeling Stress. Procedia - Social

and Behavioral Sciences, 84, 860-863.
https://doi.org/10.1016/j.sbspro.2013.06.662

Khasawneh, F. A., and Munch, E. (2016). Chatter Detection in Turning Using Persistent Homology. Mech. Syst. Signal Process. 70-71, 527–541. doi:10.1016/j.ymssp.2015.09.046

Khoshakhlagh, A.H., Sulaie, S.A., Yazdanirad, S. et al. (2024). Relationships between job stress, post-traumatic stress and musculoskeletal symptoms in firefighters and the role of job burnout and depression mediators: a bayesian network model. BMC Public Health 24, 468. https://doi.org/10.1186/s12889-024-17911-5

Li, M. Z., Ryerson, M. S., & Balakrishnan, H. (2019). Topological data analysis for aviation applications. Transportation Research Part E: Logistics and Transportation Review, 128, 149-174. https://doi.org/10.1016/j.tre.2019.05.017

Munch, E. (2017). A User's Guide to Topological Data Analysis. Journal of Learning Analytics, 4(2), 47–61. https://doi.org/10.18608/jla.2017.42.6

Munn, L. T., O'Connell, N., Huffman, C., McDonald, S., Gibbs, M., Miller, C., Danhauer, S. C., Reed, M., Mason, L., Foley, K. L., Stopyra, J., & Gesell, S. B. (2025). Job-Related Factors Associated with Burnout and Work Engagement in Emergency Nurses: Evidence to Inform Systems-Focused Interventions. Journal of Emergency Nursing, 51(2), 249-260. https://doi.org/10.1016/j.jen.2024.10.007

Neil, P. (2022). "Population estimates for the UK, England, Wales, Scotland and Northern Ireland: mid-2021". Office for National Statistics. Retrieved 14 August 2023.

Pike, J. A., Khan, A. O., Pallini, C., Thomas, S. G., Mund, M., Ries, J., Poulter, N. S., &
Styles, I. B. (2020). Topological data analysis quantifies biological nano-structure from single molecule localization microscopy. Bioinformatics, 36(5), 1614-1621.
https://doi.org/10.1093/bioinformatics/btz788

Qaiser, T., Tsang, Y.-W., Taniyama, D., Sakamoto, N., Nakane, K., Epstein, D., et al. (2019). Fast and Accurate Tumor Segmentation of Histology Images Using Persistent Homology and Deep Convolutional Features. Med. image Anal. 55, 1–14. doi:10.1016/j.media.2019.03.014

Ravishanker, N., & Chen, R. (2019). Topological Data Analysis (TDA) for Time Series. ArXiv. https://arxiv.org/abs/1909.10604

Rieck, B., Yates, T., Bock, C., Borgwardt, K., Wolf, G., Turk-Browne, N., et al. (2020). Uncovering the Topology of Time-Varying Fmri Data Using Cubical Persistence. Adv. Neural Inf. Process. Syst. 33

Rudkin, S., Rudkin, W., & Dłotko, P. (2023). On the topology of cryptocurrency markets. International Review of Financial Analysis, 89, 102759. https://doi.org/10.1016/j.irfa.2023.102759

Schmidl, S., Wenig, P., and Papenbrock. T. (2022). Anomaly detection in time series: a comprehensive evaluation. Proc. VLDB Endow. 15, 9 (May 2022), 1779–1797. https://doi.org/10.14778/3538598.3538602

Schonfeld, IS & Bianchi, R. (2016). Burnout and Depression: Two Entities or One? J Clin Psychol. 72(1):22-37. doi: 10.1002/jclp.22229. Epub 2015 Oct 9. PMID: 26451877.

Seversky, L. M., Davis, S., and Berger, M. (2016). “On Time-Series Topological Data Analysis: New Data and Opportunities,” in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, 59–67. doi:10.1109/cvprw.2016.131

Shanafelt, T., Balch, C., Bechamps, G., Russell, T., Dyrbye, L., Satele, D., et al. (2009). Burnout and career satisfaction among American surgeons. Annals of Surgery, 250, 463–471.

Skraba, P., Ovsjanikov, M., Chazal, F., and Guibas, L. (2010). "Persistence-based Segmentation of Deformable Shapes," in Computer Vision and Pattern Recognition Workshops (CVPRW), 2010 (IEEE Computer Society Conference on), 45–52. doi:10.1109/cvprw.2010.5543285

Smith, A. D., Dłotko, P., & Zavala, V. M. (2021). Topological data analysis: Concepts, computation, and applications in chemical engineering. Computers & Chemical Engineering, 146, 107202. https://doi.org/10.1016/j.compchemeng.2020.107202

Turner, K., Mukherjee, S., and Boyer, D. M. (2014). Persistent Homology Transform for Modeling Shapes and Surfaces. Inf. Inference 3 (4), 310–344. doi:10.1093/imaiai/iau011

Umeda, Y. (2017). Time Series Classification via Topological Data Analysis. Trans. Jpn. Soc. Artif. Intell. 32 (3), D–G72_1. doi:10.1527/tjsai.d-g72

Wikipedia (2025). Budget of the United Kingdom. https://en.wikipedia.org/wiki/Budget_of_the_United_Kingdom#:~:text=The%20UK%20fiscal%20year%20ends,referred%20to%20as%202021%E2%80%9322.

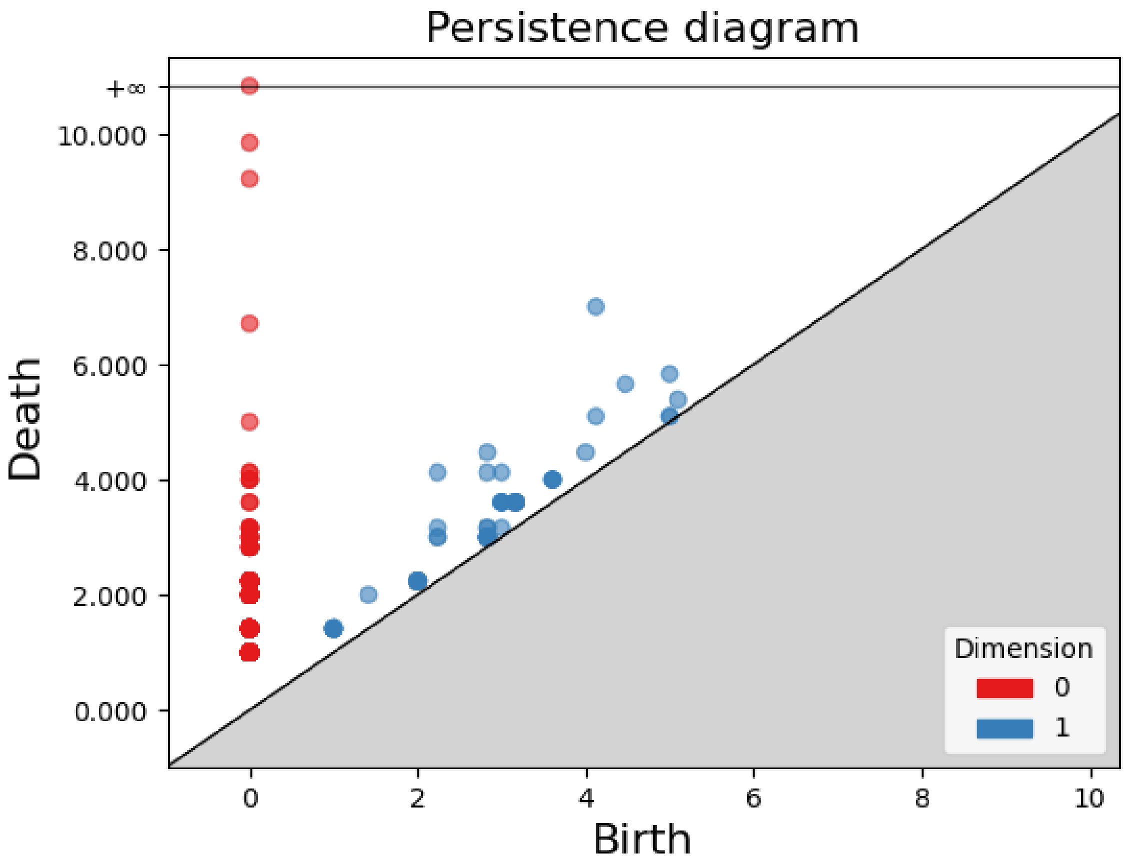
Wilson, S., Daar, D., Sinno, S., & Levine, S. (2018). Public interest in breast augmentation: Analysis and implications of google trends data. Aesthetic Plastic Surgery, 42(3), 648–655.

Y. Gu, J. Guo, C. He and D. Liu. (2022). The evolution pattern of world trade network based on topological data analysis. 14th International Conference on Measuring Technology and Mechatronics Automation (ICMTMA), Changsha, China, 2022, pp. 1119-1122, doi: 10.1109/ICMTMA54903.2022.00225.

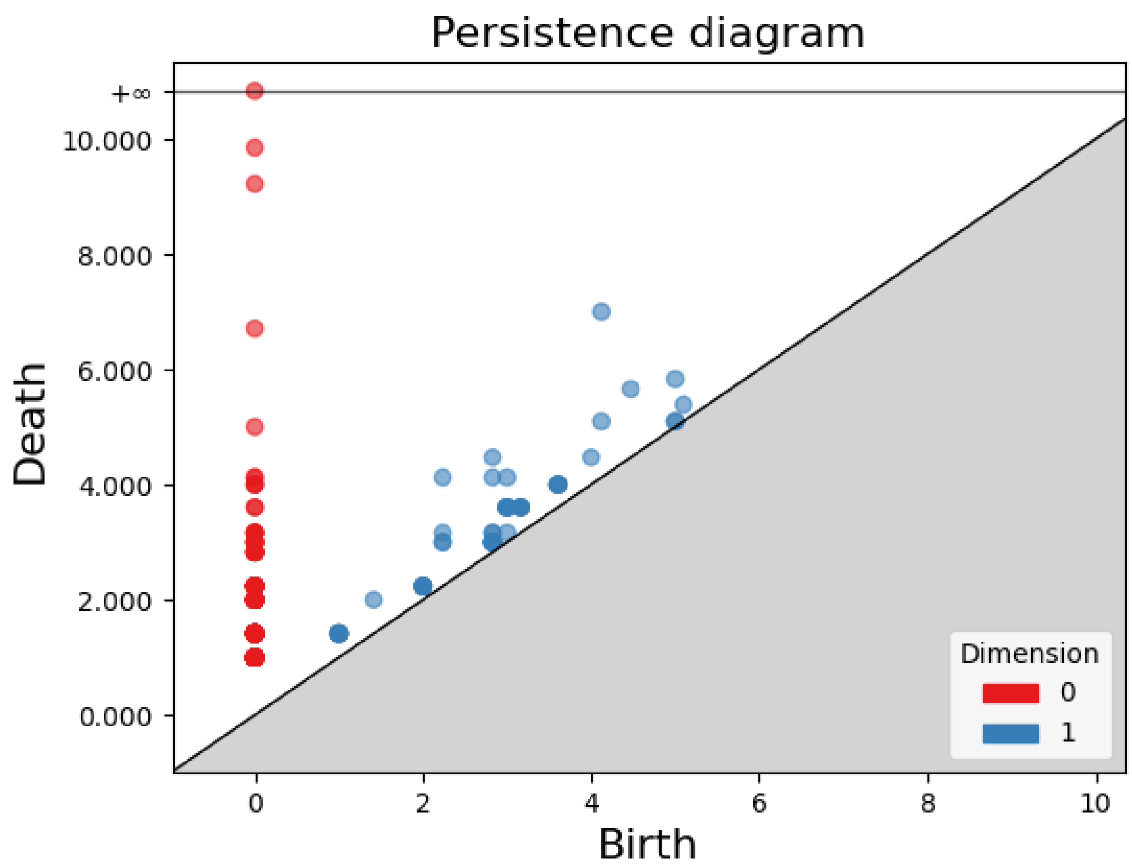
APPENDIX

Persistence diagrams:

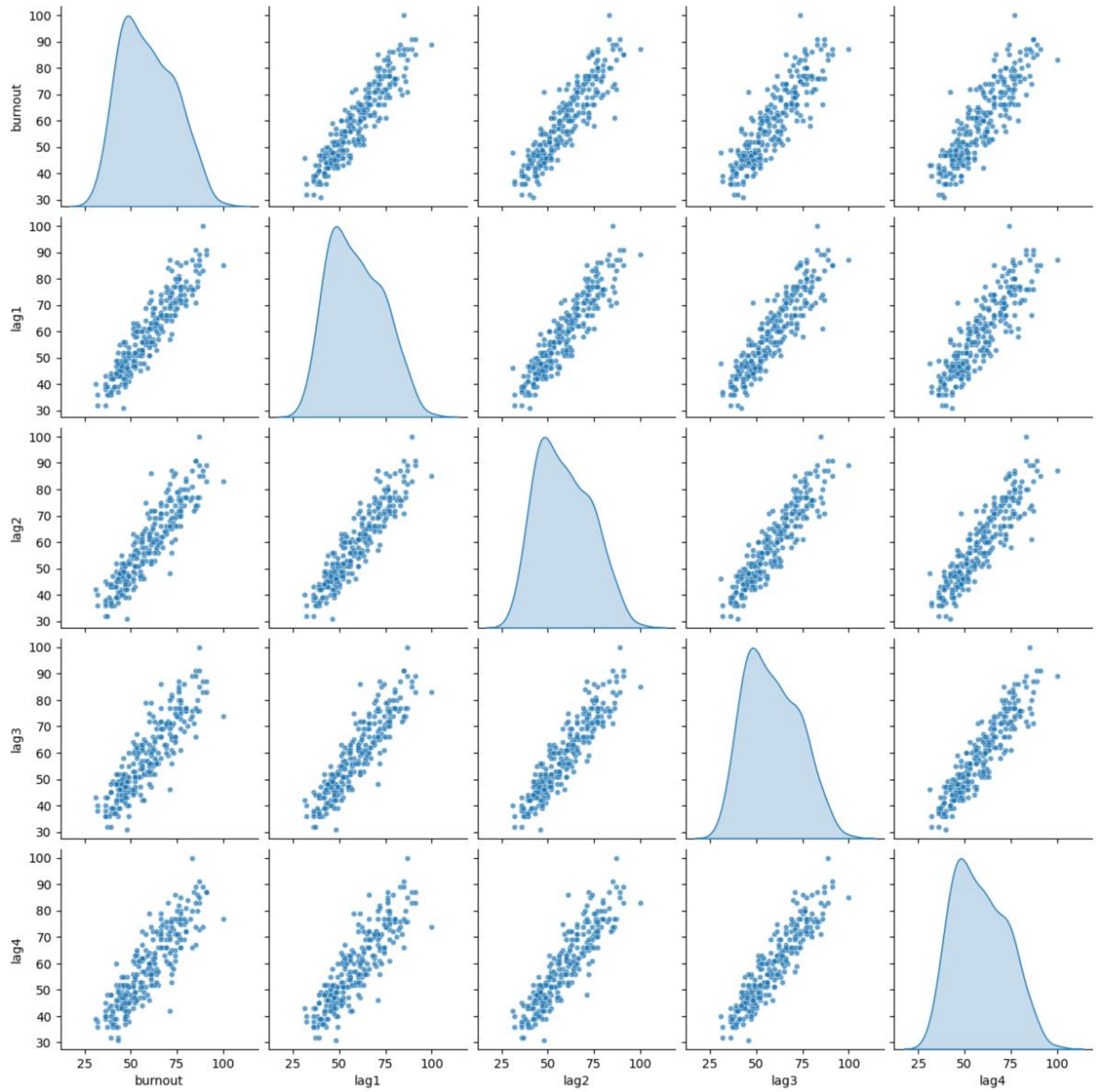
Cloud A



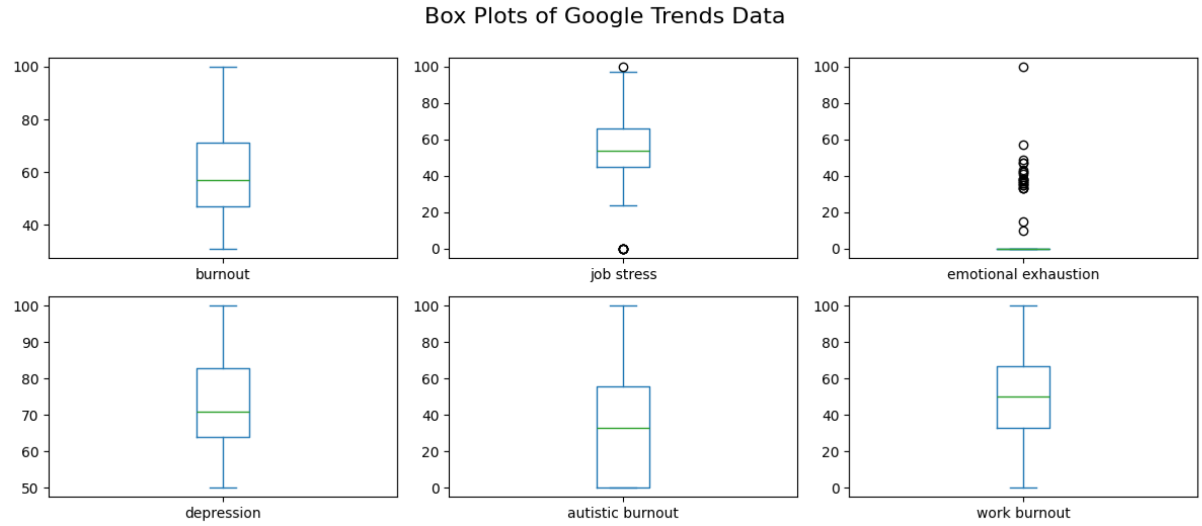
Cloud C

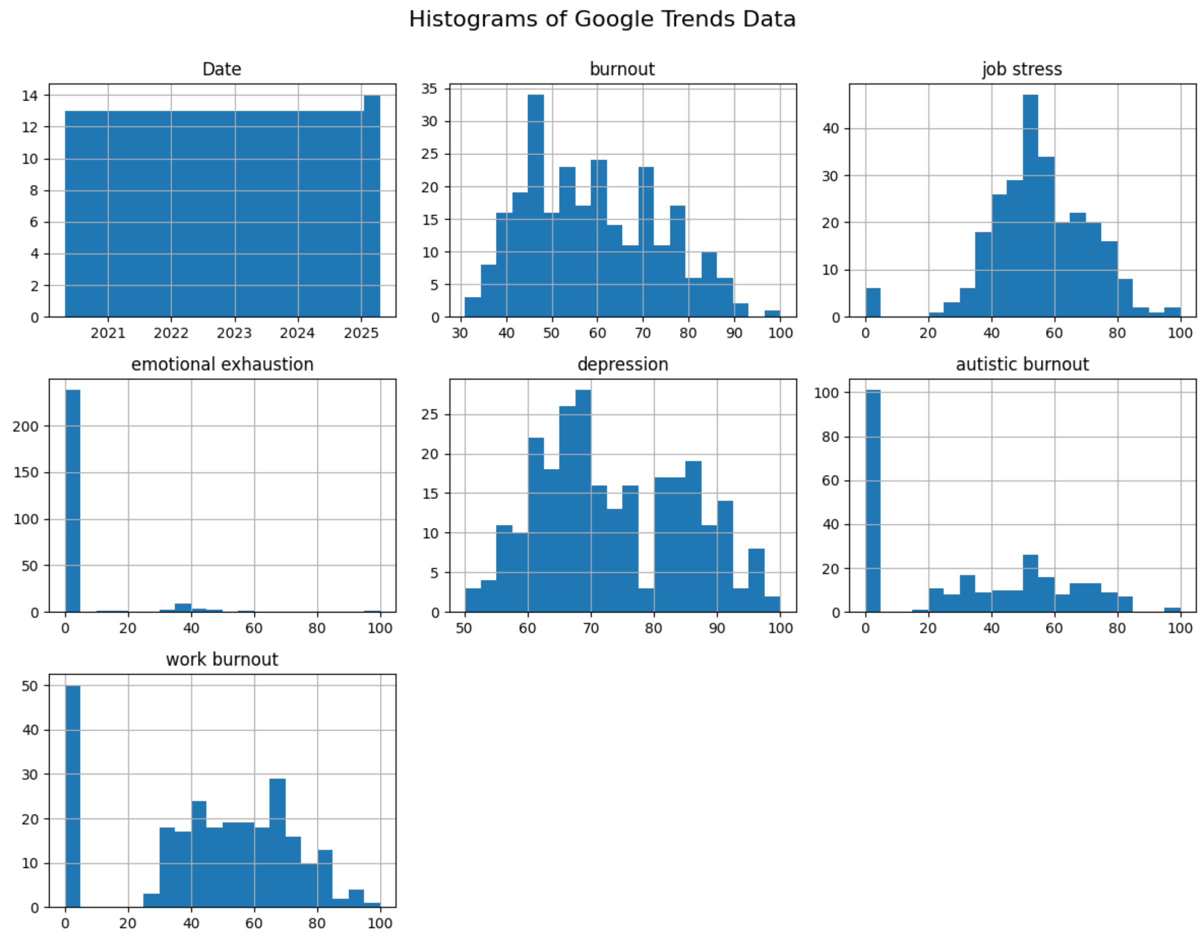


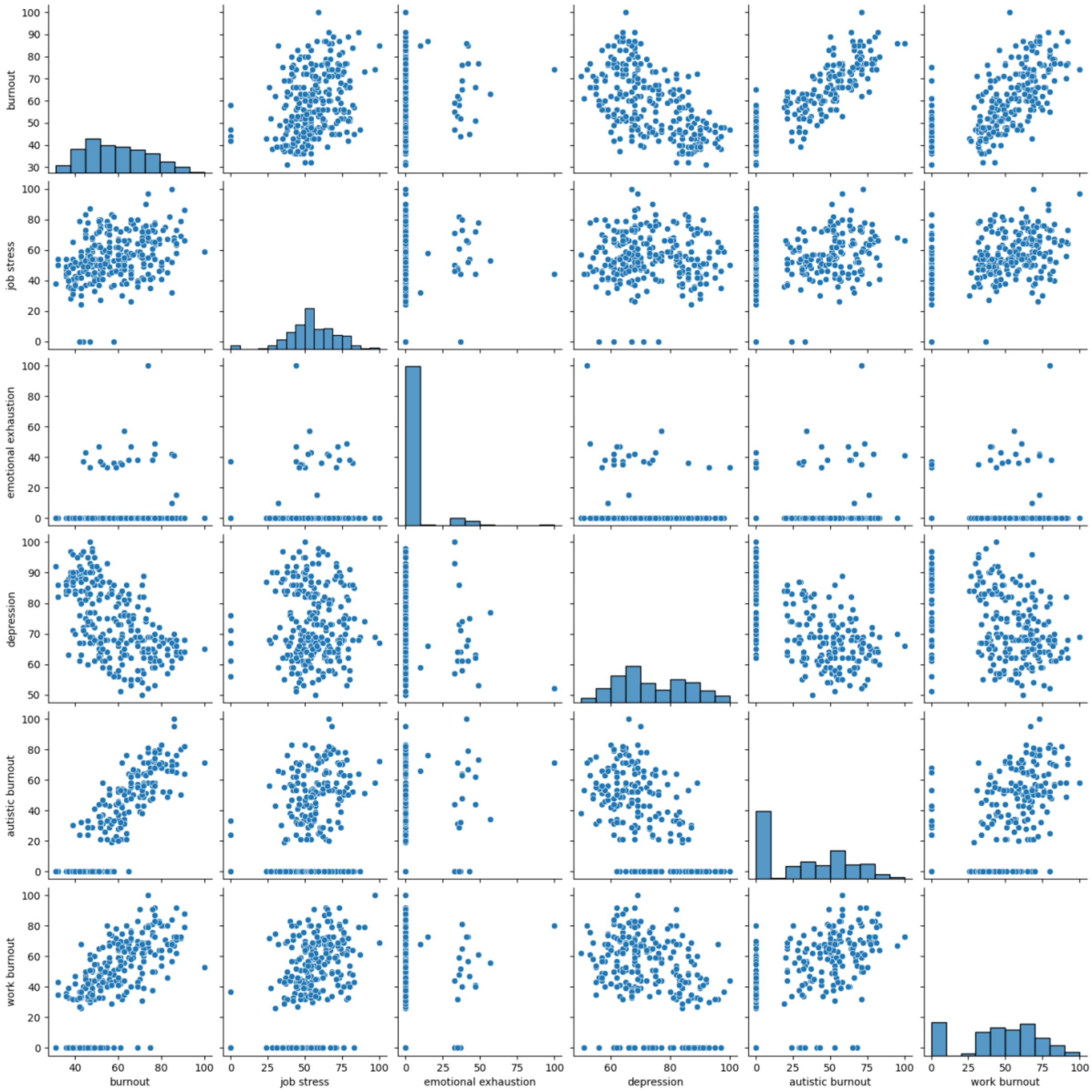
Pairplot for Cloud B (+4 lag)



Additional EDA diagrams:


Box Plots of Google Trends Data
burnout
job stress
emotional exhaustion
depression
autistic burnout
work burnout


Histograms of Google Trends Data
Date
burnout
job stress
emotional exhaustion
depression
autistic burnout
work burnout


burnout
job stress
emotional exhaustion
depression
autistic burnout
work burnout

Code:

Downloading all the Libraries

In [2]:
```
!pip install gudhi
!pip install urllib3==1.26.16
!pip install Requests
!pip install math
!pip install POT
!pip install pytrends
import pandas as pd
from functools import reduce
import math
import numpy as np
from pytrends.request import TrendReq
import gudhi as gd
import gudhi.wasserstein
import gudhi.hera
import seaborn as sb
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
from scipy.stats import skew, kurtosis
from scipy.stats import spearmanr
import matplotlib as mpl
```

```
Collecting gudhi
  Downloading gudhi-3.11.0-cp311-cp311-manylinux_2_28_x86_64.whl.metadata
(1.6 kB)
Requirement already satisfied: numpy>=1.15.0 in /usr/local/lib/python3.11/d
ist-packages (from gudhi) (2.0.2)
Downloading gudhi-3.11.0-cp311-cp311-manylinux_2_28_x86_64.whl (4.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 4.2/4.2 MB 33.3 MB/s eta 0:00:0
0
Installing collected packages: gudhi
Successfully installed gudhi-3.11.0
Collecting urllib3==1.26.16
  Downloading urllib3-1.26.16-py2.py3-none-any.whl.metadata (48 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 48.4/48.4 kB 1.8 MB/s eta 0:0
0:00
Downloading urllib3-1.26.16-py2.py3-none-any.whl (143 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 143.1/143.1 kB 5.0 MB/s eta 0:0
0:00
Installing collected packages: urllib3
  Attempting uninstall: urllib3
    Found existing installation: urllib3 2.4.0
    Uninstalling urllib3-2.4.0:
      Successfully uninstalled urllib3-2.4.0
Successfully installed urllib3-1.26.16
Requirement already satisfied: Requests in /usr/local/lib/python3.11/dist-p
ackages (2.32.3)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/p
ython3.11/dist-packages (from Requests) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/di
st-packages (from Requests) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python
3.11/dist-packages (from Requests) (1.26.16)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python
3.11/dist-packages (from Requests) (2025.4.26)
ERROR: Could not find a version that satisfies the requirement math (from v
ersions: none)
ERROR: No matching distribution found for math
Collecting POT
  Downloading POT-0.9.5-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86
_64.whl.metadata (34 kB)
Requirement already satisfied: numpy>=1.16 in /usr/local/lib/python3.11/dis
t-packages (from POT) (2.0.2)
Requirement already satisfied: scipy>=1.6 in /usr/local/lib/python3.11/dist
-packages (from POT) (1.15.2)
Downloading POT-0.9.5-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_6
4.whl (897 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 897.5/897.5 kB 14.4 MB/s eta 0:
00:00
Installing collected packages: POT
Successfully installed POT-0.9.5
Collecting pytrends
  Downloading pytrends-4.9.2-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: requests>=2.0 in /usr/local/lib/python3.11/d
ist-packages (from pytrends) (2.32.3)
Requirement already satisfied: pandas>=0.25 in /usr/local/lib/python3.11/di
st-packages (from pytrends) (2.2.2)
Requirement already satisfied: lxml in /usr/local/lib/python3.11/dist-packa
ges (from pytrends) (5.4.0)
Requirement already satisfied: numpy>=1.23.2 in /usr/local/lib/python3.11/d
ist-packages (from pandas>=0.25->pytrends) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/pyt
hon3.11/dist-packages (from pandas>=0.25->pytrends) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/di
st-packages (from pandas>=0.25->pytrends) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/
```

```
dist-packages (from pandas>=0.25->pytrends) (2025.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/p
ython3.11/dist-packages (from requests>=2.0->pytrends) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/di
st-packages (from requests>=2.0->pytrends) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python
3.11/dist-packages (from requests>=2.0->pytrends) (1.26.16)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python
3.11/dist-packages (from requests>=2.0->pytrends) (2025.4.26)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-p
ackages (from python-dateutil>=2.8.2->pandas>=0.25->pytrends) (1.17.0)
Downloading pytrends-4.9.2-py3-none-any.whl (15 kB)
Installing collected packages: pytrends
Successfully installed pytrends-4.9.2
```

In [5]:

```python
# Load dataset
data = pd.read_excel("TDA.xlsx")

print(data.head())
```

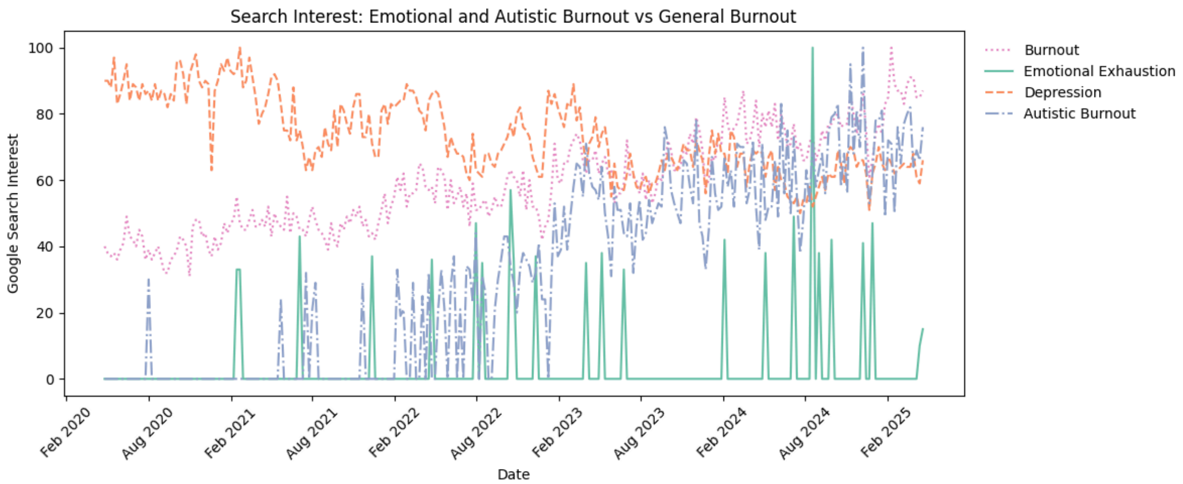
```
        Date  burnout  job stress  emotional exhaustion  depression  \
0 2020-04-26       40          43                     0          90
1 2020-05-03       38          28                     0          90
2 2020-05-10       37          36                     0          88
3 2020-05-17       38          35                     0          97
4 2020-05-24       36          43                     0          83

   autistic burnout  work burnout
0                 0             0
1                 0             0
2                 0            33
3                 0             0
4                 0            36
```

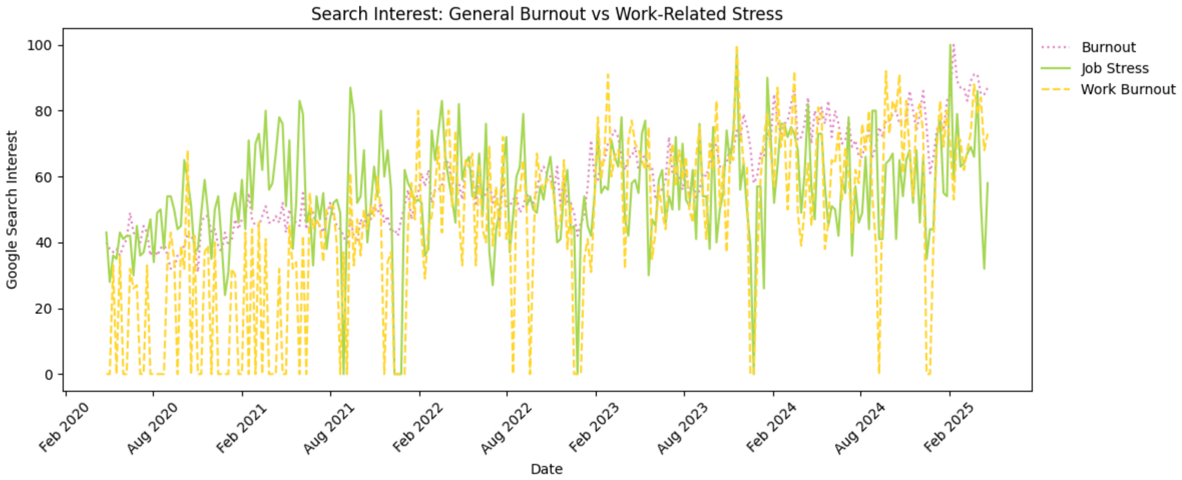
Creating Timeseries Plots

In [6]:

```python
distinct_colors = sb.color_palette("Set2")

# Assign colors to each variable
colors = {
    "emotional exhaustion": distinct_colors[0],
    "depression": distinct_colors[1],
    "autistic burnout": distinct_colors[2],
    "burnout": distinct_colors[3],
    "job stress": distinct_colors[4],
    "work burnout": distinct_colors[5],
}

# Assign line styles
linestyles = {
    "emotional exhaustion": "-",
    "depression": "--",
    "autistic burnout": "-.",
    "burnout": ":",
    "job stress": "-",
    "work burnout": "--",
}

# Plot 1 (Time Series for Burnout and Mental Health Terms)
plt.figure(figsize=(12, 5))

sb.lineplot(x="Date", y="burnout", data=data, label="Burnout",
            color=colors["burnout"], linestyle=linestyles["burnout"])
sb.lineplot(x="Date", y="emotional exhaustion", data=data, label="Emotional
```

```
             color=colors["emotional exhaustion"], linestyle=linestyles["emot
sb.lineplot(x="Date", y="depression", data=data, label="Depression",
             color=colors["depression"], linestyle=linestyles["depression"])
sb.lineplot(x="Date", y="autistic burnout", data=data, label="Autistic Burno
             color=colors["autistic burnout"], linestyle=linestyles["autistic


plt.gca().xaxis.set_major_locator(mdates.MonthLocator(interval=6))
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%b %Y'))
plt.xticks(rotation=45)

plt.ylabel("Google Search Interest")
plt.title("Search Interest: Emotional and Autistic Burnout vs General Burnou
plt.legend(loc='upper center', bbox_to_anchor=(1.13, 1), frameon=False)
plt.tight_layout()
plt.show()

# Plot 2 (Time Series for Burnout and Occupational Terms)
plt.figure(figsize=(12, 5))

sb.lineplot(x="Date", y="burnout", data=data, label="Burnout",
             color=colors["burnout"], linestyle=linestyles["burnout"])
sb.lineplot(x="Date", y="job stress", data=data, label="Job Stress",
             color=colors["job stress"], linestyle=linestyles["job stress"])
sb.lineplot(x="Date", y="work burnout", data=data, label="Work Burnout",
             color=colors["work burnout"], linestyle=linestyles["work burnou

plt.gca().xaxis.set_major_locator(mdates.MonthLocator(interval=6))
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%b %Y'))
plt.xticks(rotation=45)

plt.ylabel("Google Search Interest")
plt.title("Search Interest: General Burnout vs Work-Related Stress")
plt.legend(loc='upper center', bbox_to_anchor=(1.08,1), frameon=False)
plt.tight_layout()
plt.show()
```

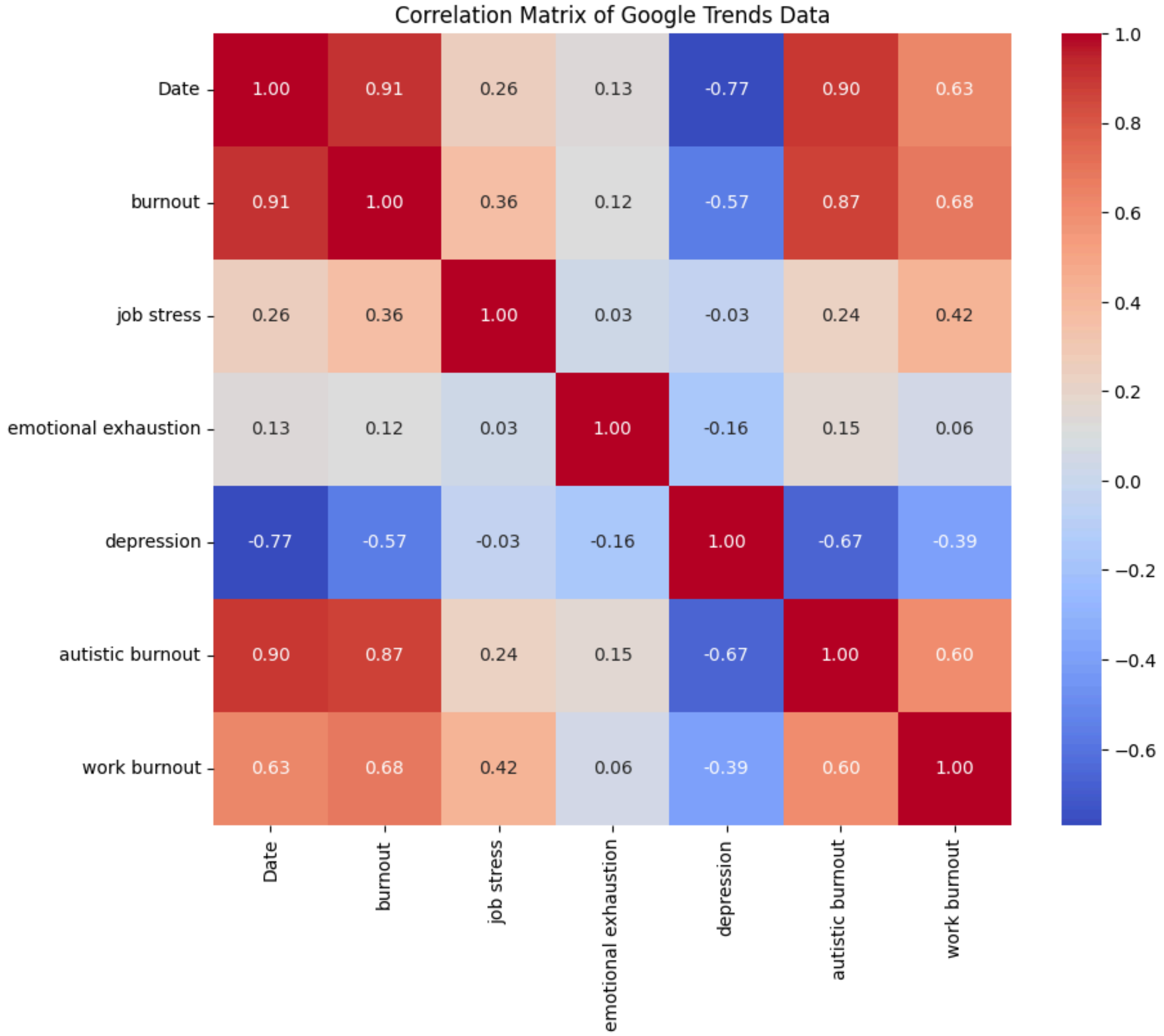


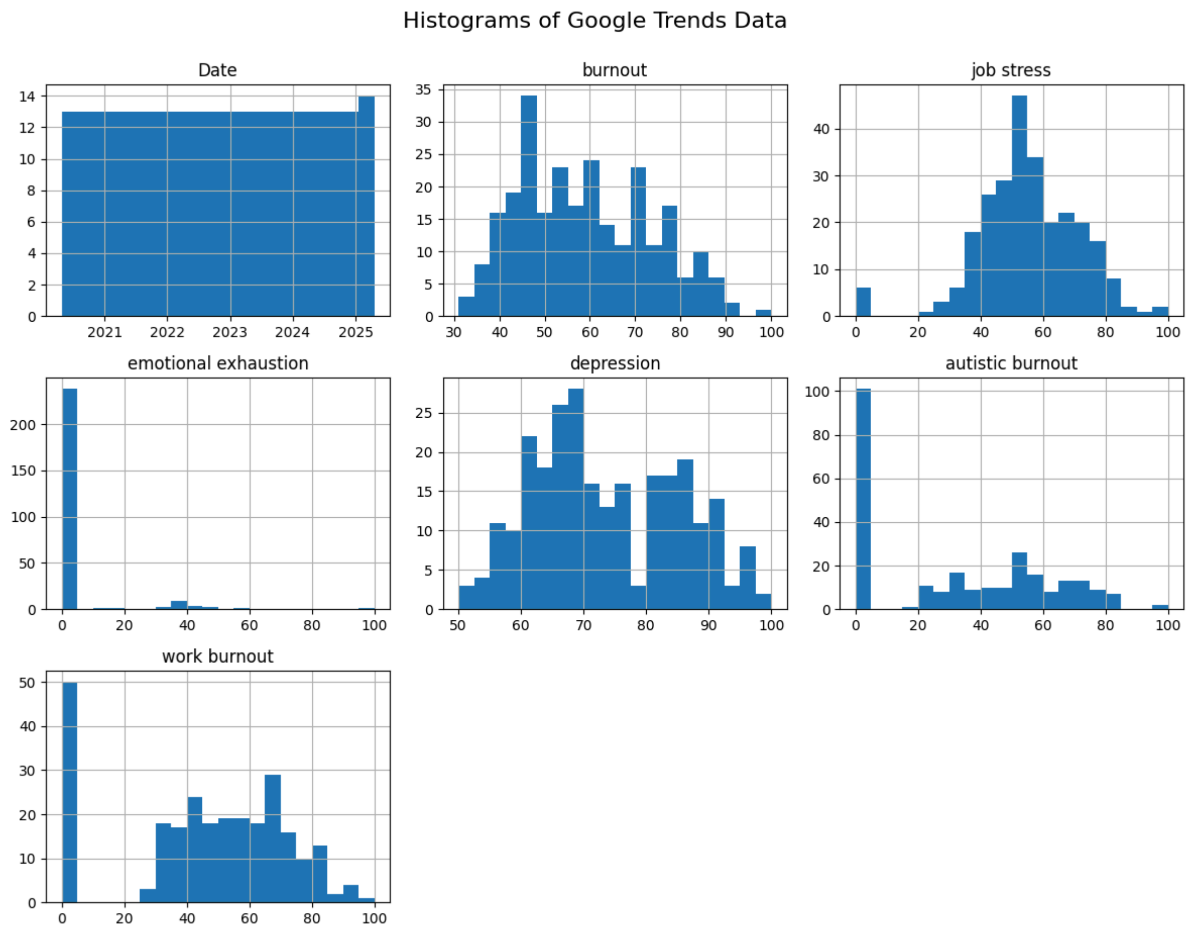


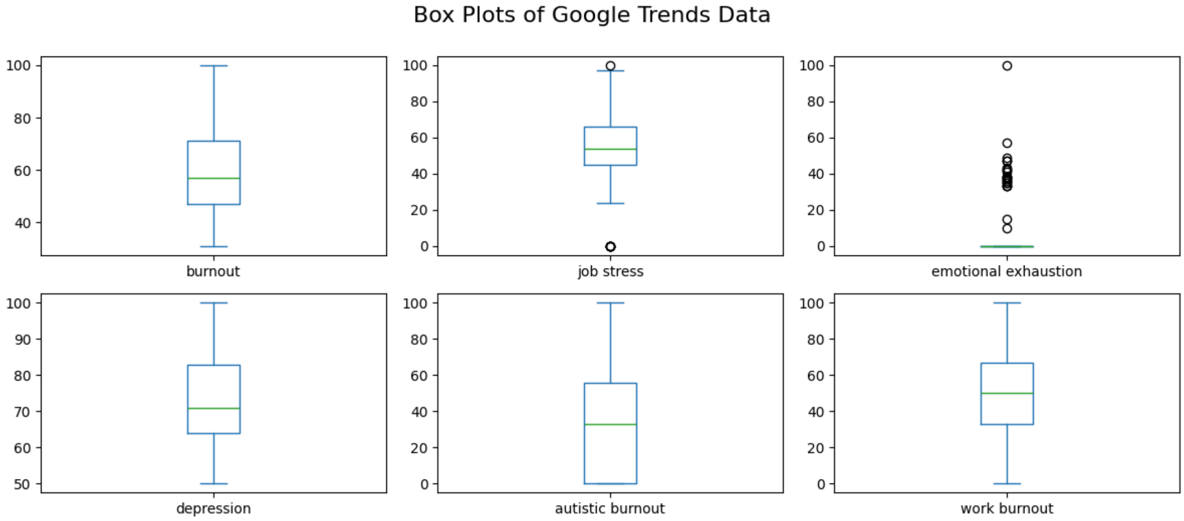
Summary Statistics and EDA

In [ ]:
```
# Summary Statistics
print(data.describe())

# Data Types
print(data.dtypes)

# Missing Values
print(data.isnull().sum())

# Correlation Matrix
correlation_matrix = data.corr()
print(correlation_matrix)

# Heatmap
plt.figure(figsize=(10, 8))
sb.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix of Google Trends Data')
plt.show()

# Histogram
data.hist(figsize=(12, 10), bins=20)
plt.suptitle("Histograms of Google Trends Data", y=0.95, fontsize=16)
plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.show()

# Box plot
data.plot(kind='box', subplots=True, layout=(3, 3), sharex=False, sharey=Fa
plt.suptitle('Box Plots of Google Trends Data', y=0.95, fontsize=16)
plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.show()

# Skewness and Kurtosis
for col in data.columns[1:]:  # Exclude the 'Date' column
    print(f"Skewness of {col}: {skew(data[col])}")
    print(f"Kurtosis of {col}: {kurtosis(data[col])}")

# Pairplot
sb.pairplot(data)
plt.show()
```

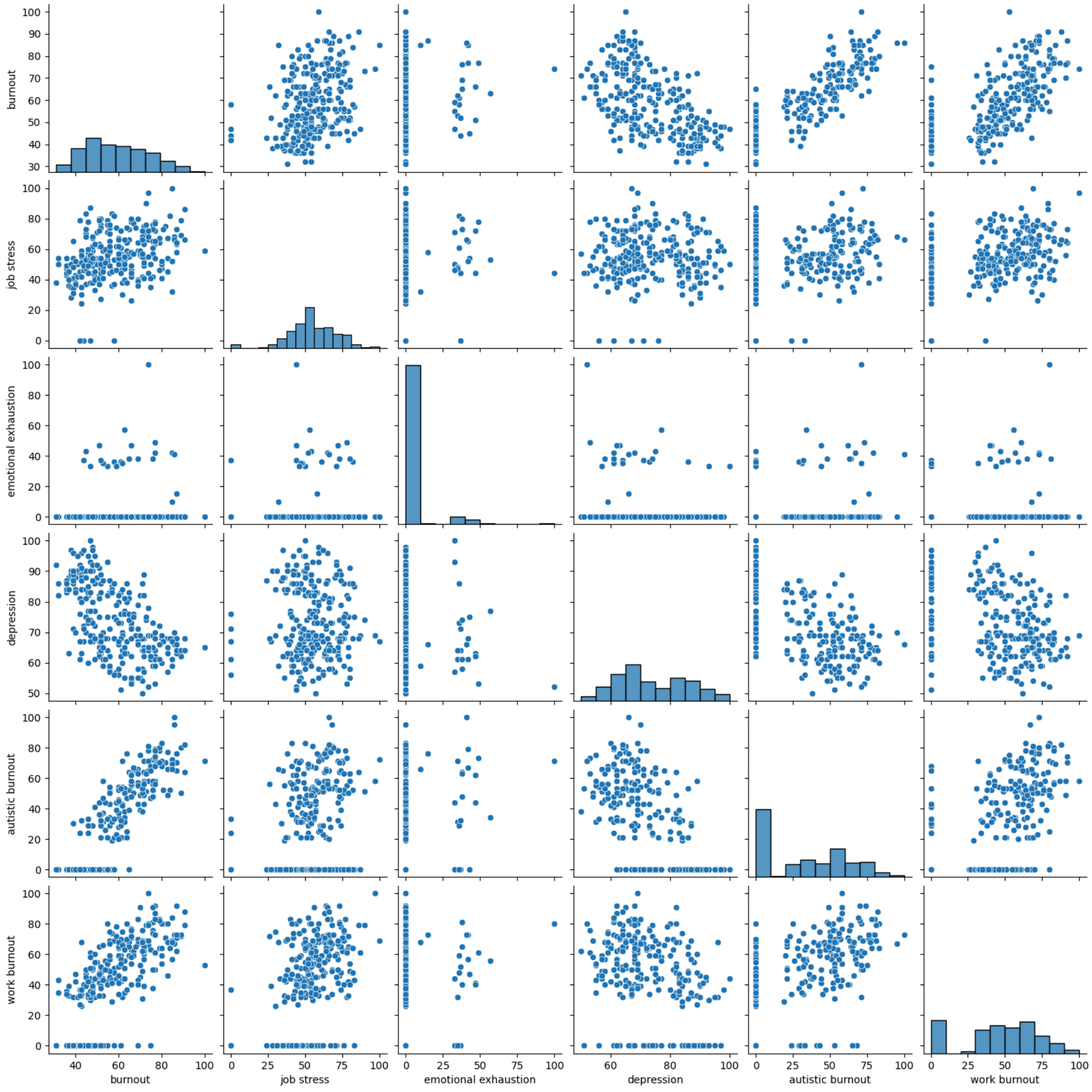
```
                      Date     burnout  job stress  emotional exhaustion  \
count                  261  261.000000  261.000000            261.000000
mean   2022-10-23 00:00:00   58.881226   54.842912              3.532567
min    2020-04-26 00:00:00   31.000000    0.000000              0.000000
25%    2021-07-25 00:00:00   47.000000   45.000000              0.000000
50%    2022-10-23 00:00:00   57.000000   54.000000              0.000000
75%    2024-01-21 00:00:00   71.000000   66.000000              0.000000
max    2025-04-20 00:00:00  100.000000  100.000000            100.000000
std                    NaN   14.555903   16.198357             12.337088

       depression  autistic burnout  work burnout
count  261.000000        261.000000    261.000000
mean    73.413793         31.781609     45.697318
min     50.000000          0.000000      0.000000
25%     64.000000          0.000000     33.000000
50%     71.000000         33.000000     50.000000
75%     83.000000         56.000000     67.000000
max    100.000000        100.000000    100.000000
std     11.639342         28.864441     26.622799
Date                    datetime64[ns]
burnout                          int64
job stress                       int64
emotional exhaustion             int64
depression                       int64
autistic burnout                 int64
work burnout                     int64
dtype: object
Date                    0
burnout                 0
job stress              0
emotional exhaustion    0
depression              0
autistic burnout        0
work burnout            0
dtype: int64
                          Date   burnout  job stress  emotional exhaustion
\
Date                  1.000000  0.909291    0.258684              0.131346
burnout               0.909291  1.000000    0.356704              0.118837
job stress            0.258684  0.356704    1.000000              0.028135
emotional exhaustion  0.131346  0.118837    0.028135              1.000000
depression           -0.770132 -0.566344   -0.026908             -0.161231
autistic burnout      0.901587  0.868698    0.236796              0.153654
work burnout          0.628846  0.678713    0.420487              0.058223

                      depression  autistic burnout  work burnout
Date                   -0.770132          0.901587      0.628846
burnout                -0.566344          0.868698      0.678713
job stress             -0.026908          0.236796      0.420487
emotional exhaustion   -0.161231          0.153654      0.058223
depression              1.000000         -0.665749     -0.394820
autistic burnout       -0.665749          1.000000      0.603349
work burnout           -0.394820          0.603349      1.000000
```


Correlation Matrix of Google Trends Data
Date
burnout
job stress
emotional exhaustion
depression
autistic burnout
work burnout
1.00 0.91 0.26 0.13 -0.77 0.90 0.63
0.91 1.00 0.36 0.12 -0.57 0.87 0.68
0.26 0.36 1.00 0.03 -0.03 0.24 0.42
0.13 0.12 0.03 1.00 -0.16 0.15 0.06
-0.77 -0.57 -0.03 -0.16 1.00 -0.67 -0.39
0.90 0.87 0.24 0.15 -0.67 1.00 0.60
0.63 0.68 0.42 0.06 -0.39 0.60 1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6

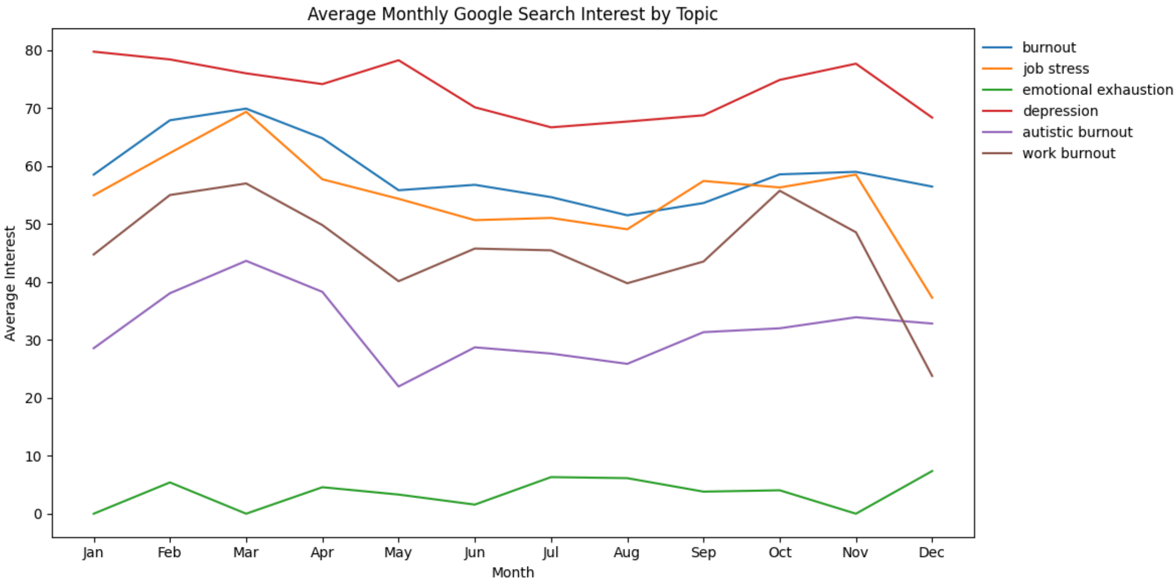

Histograms of Google Trends Data
Date
burnout
job stress
emotional exhaustion
depression
autistic burnout
work burnout




```
Skewness of burnout: 0.33187751799997417
Kurtosis of burnout: -0.7603692470173926
Skewness of job stress: -0.5254021414146458
Kurtosis of job stress: 1.6479478320330925
Skewness of emotional exhaustion: 3.967692984233465
Kurtosis of emotional exhaustion: 18.367722516104035
Skewness of depression: 0.24326601736323072
Kurtosis of depression: -0.9622439713177435
Skewness of autistic burnout: 0.1821398543174791
Kurtosis of autistic burnout: -1.3767075237630395
Skewness of work burnout: -0.5193794356319414
Kurtosis of work burnout: -0.7056409151046164
```

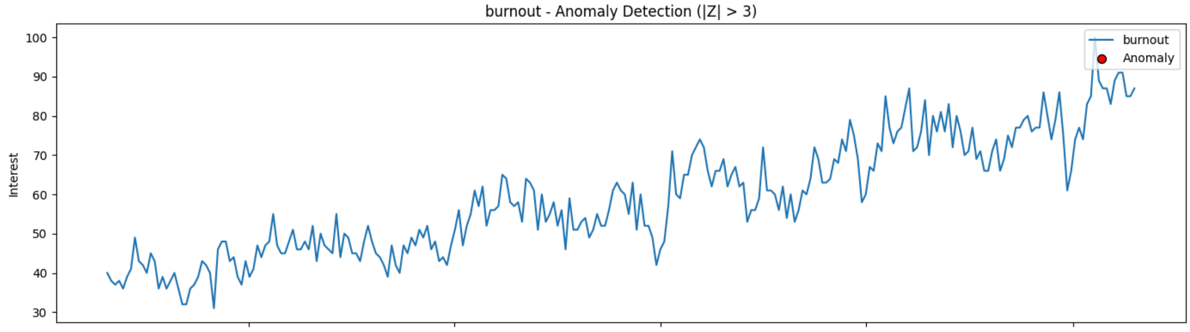


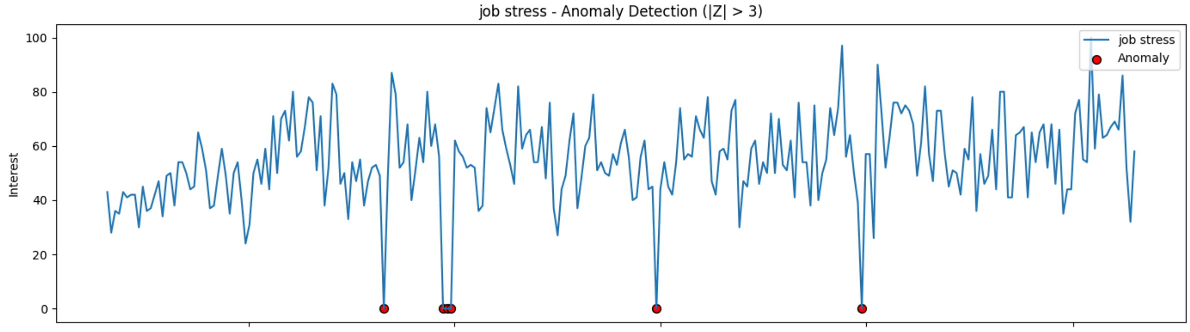
In [ ]:
```
# Ensure 'Date' is in datetime format
data['Date'] = pd.to_datetime(data['Date'])

# Extract the month as a number (1=Jan, 12=Dec)
data['Month'] = data['Date'].dt.month

# Group by month and calculate mean search interest for each variable (exclu
monthly_avg = data.drop(columns='Date').groupby('Month').mean()

# Plot
plt.figure(figsize=(12, 6))
for column in monthly_avg.columns:
    plt.plot(monthly_avg.index, monthly_avg[column], label=column)

plt.xticks(ticks=range(1, 13), labels=['Jan', 'Feb', 'Mar', 'Apr', 'May', 'J
                                       'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'D
plt.title("Average Monthly Google Search Interest by Topic")
plt.xlabel("Month")
plt.ylabel("Average Interest")
plt.legend(loc='upper right', bbox_to_anchor=(1.23, 1), frameon=False)
plt.tight_layout()
plt.show()
```

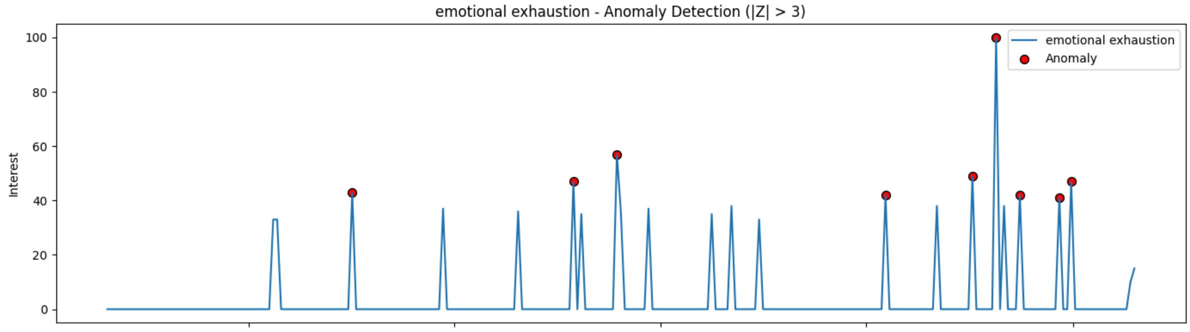


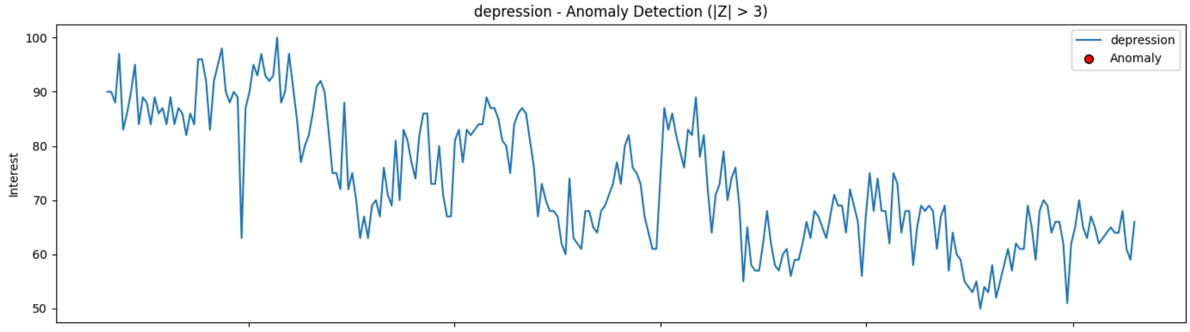
In [ ]:

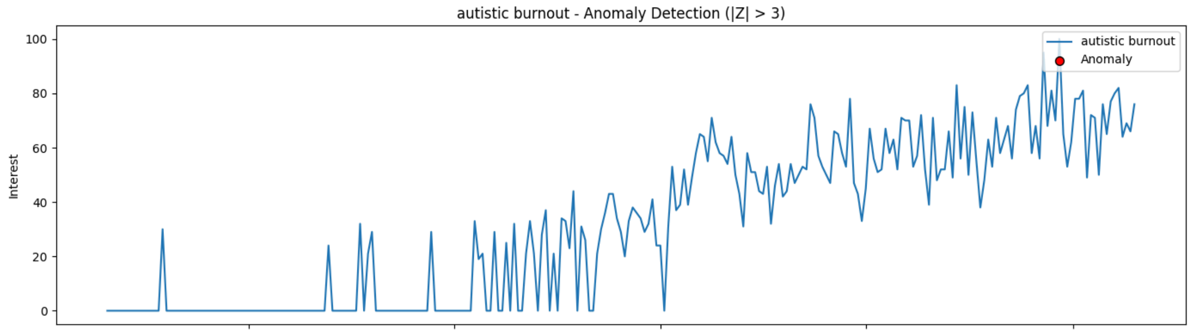
```
# Drop Date column and compute Z-scores
search_data = data.drop(columns='Date')
z_scores = (search_data - search_data.mean()) / search_data.std()
threshold = 3
anomalies = z_scores.abs() > threshold

# Set up subplots
num_vars = len(search_data.columns)
fig, axes = plt.subplots(num_vars, 1, figsize=(14, 4 * num_vars), sharex=Tru

# Loop through each variable
for i, col in enumerate(search_data.columns):
    ax = axes[i]

    # Plot the time series
    ax.plot(data['Date'], data[col], label=col, color='tab:blue')

    # Plot anomalies
    ax.scatter(
        data['Date'][anomalies[col]],
        data[col][anomalies[col]],
        color='red', s=50, label='Anomaly', marker='o', edgecolors='black'
    )

    ax.set_title(f"{col} - Anomaly Detection (|Z| > {threshold})")
    ax.set_ylabel("Interest")
    ax.legend(loc='upper right')

# Set common x-label
plt.xlabel("Date")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

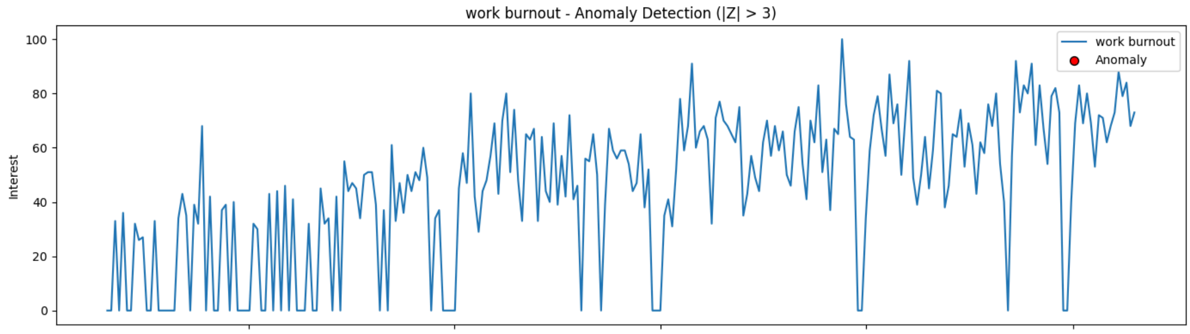

burnout - Anomaly Detection (|Z| > 3)
Interest
burnout
Anomaly

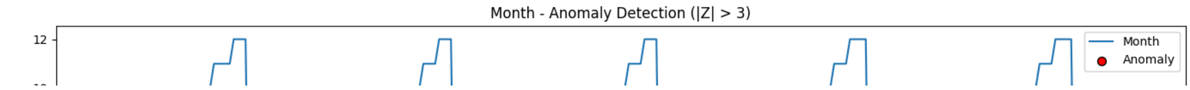

job stress - Anomaly Detection (|Z| > 3)
Interest
job stress
Anomaly

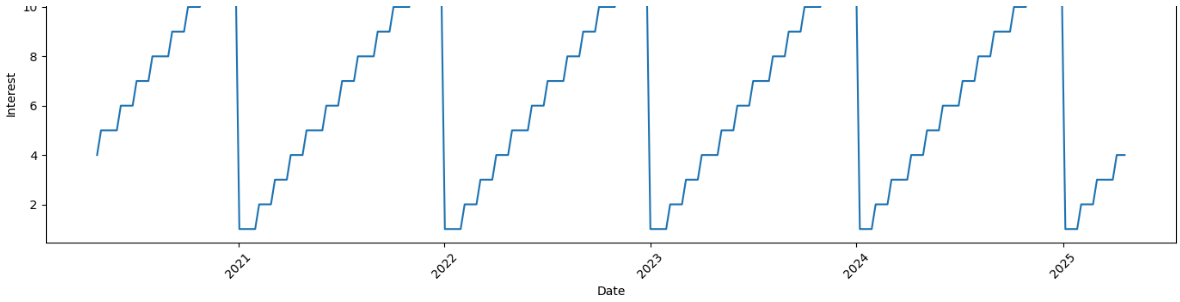

emotional exhaustion - Anomaly Detection (|Z| > 3)
Interest
emotional exhaustion
Anomaly


depression - Anomaly Detection (|Z| > 3)
Interest
depression
Anomaly

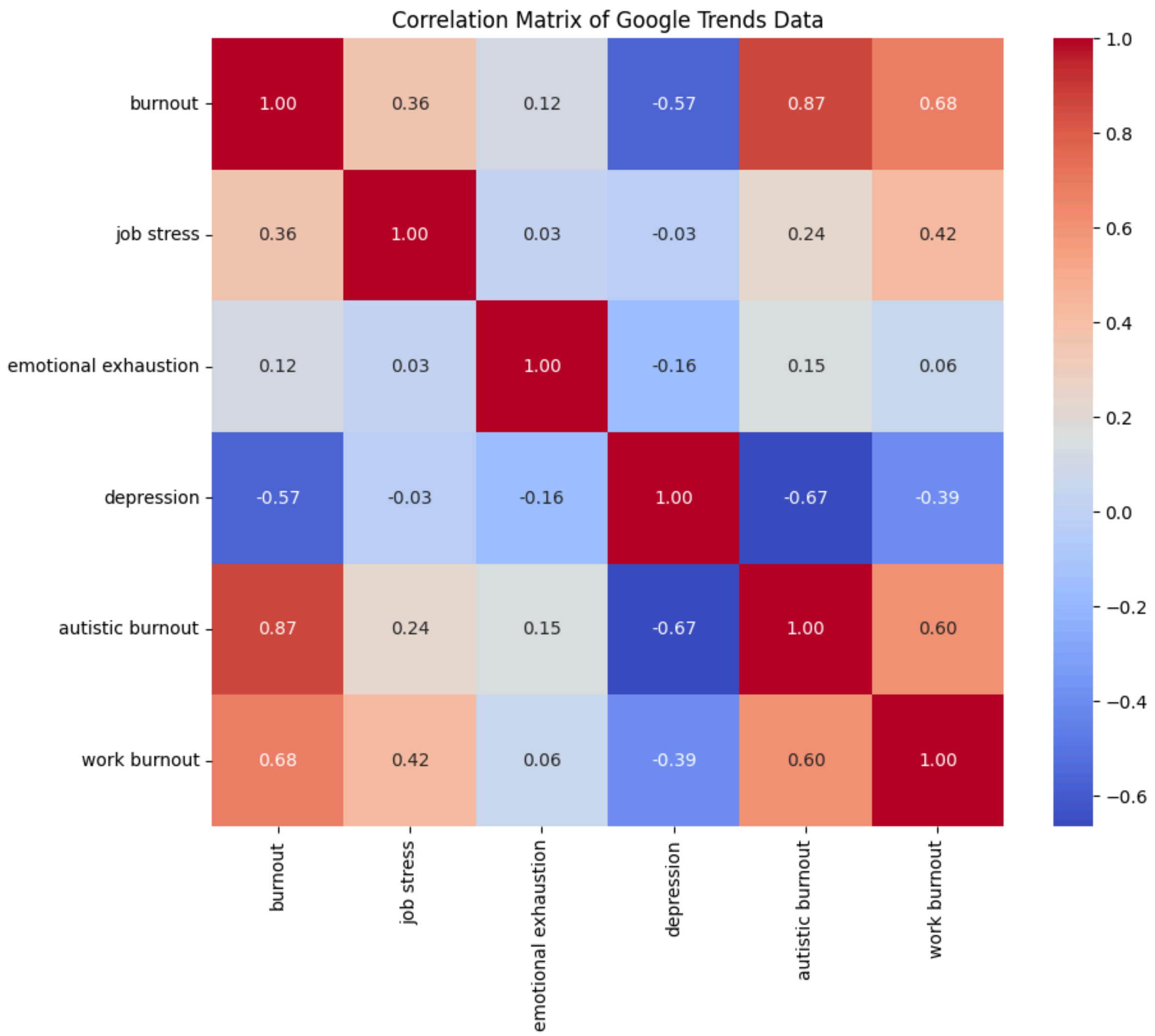

autistic burnout - Anomaly Detection (|Z| > 3)
Interest
autistic burnout
Anomaly


work burnout - Anomaly Detection (|Z| > 3)
Interest
work burnout
Anomaly


Month - Anomaly Detection (|Z| > 3)
Month
Anomaly



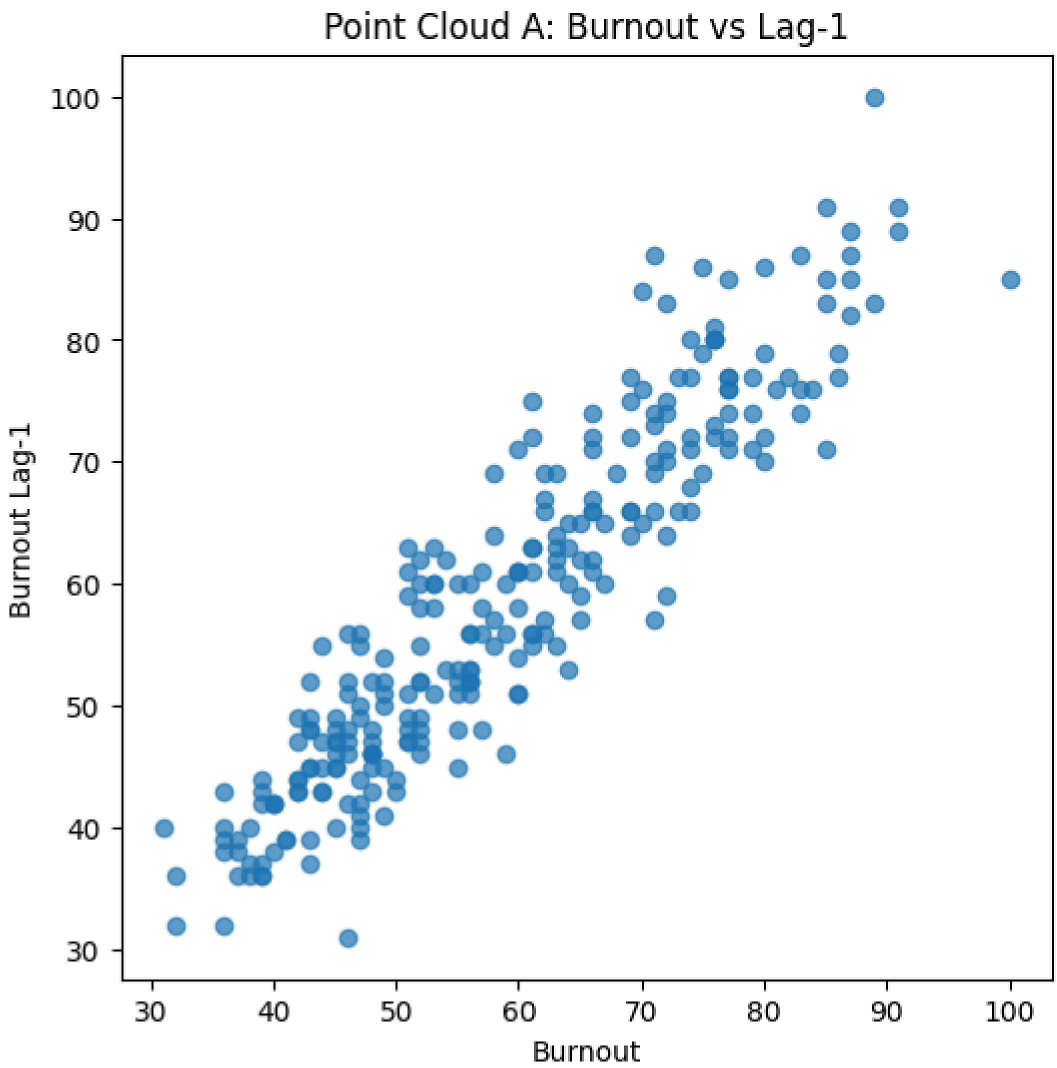
In [16]:
```
# Create a new DataFrame without the 'Date' column
data_no_date = data.drop(columns=['Date'])

# Calculate the correlation matrix
correlation_matrix_no_date = data_no_date.corr()

# Visualize the correlation matrix
plt.figure(figsize=(10, 8))
sb.heatmap(correlation_matrix_no_date, annot=True, cmap='coolwarm', fmt=".2
plt.title('Correlation Matrix of Google Trends Data')
plt.show()
```



Point Cloud A

In [18]:
```
# Create lagged series (Lag-1)
data['burnout_lag1'] = data['burnout'].shift(1)

# Drop any rows with NaN caused by shifting
point_cloud = data.dropna(subset=['burnout', 'burnout_lag1'])
```

```
# Convert to NumPy array for TDA tools like Gudhi
cloud_A = point_cloud[['burnout', 'burnout_lag1']].to_numpy()

# Visualize the point cloud
plt.figure(figsize=(6, 6))
plt.scatter(cloud_A[:, 0], cloud_A[:, 1], alpha=0.7)
plt.title('Point Cloud A: Burnout vs Lag-1')
plt.xlabel('Burnout')
plt.ylabel('Burnout Lag-1')
plt.show()
```

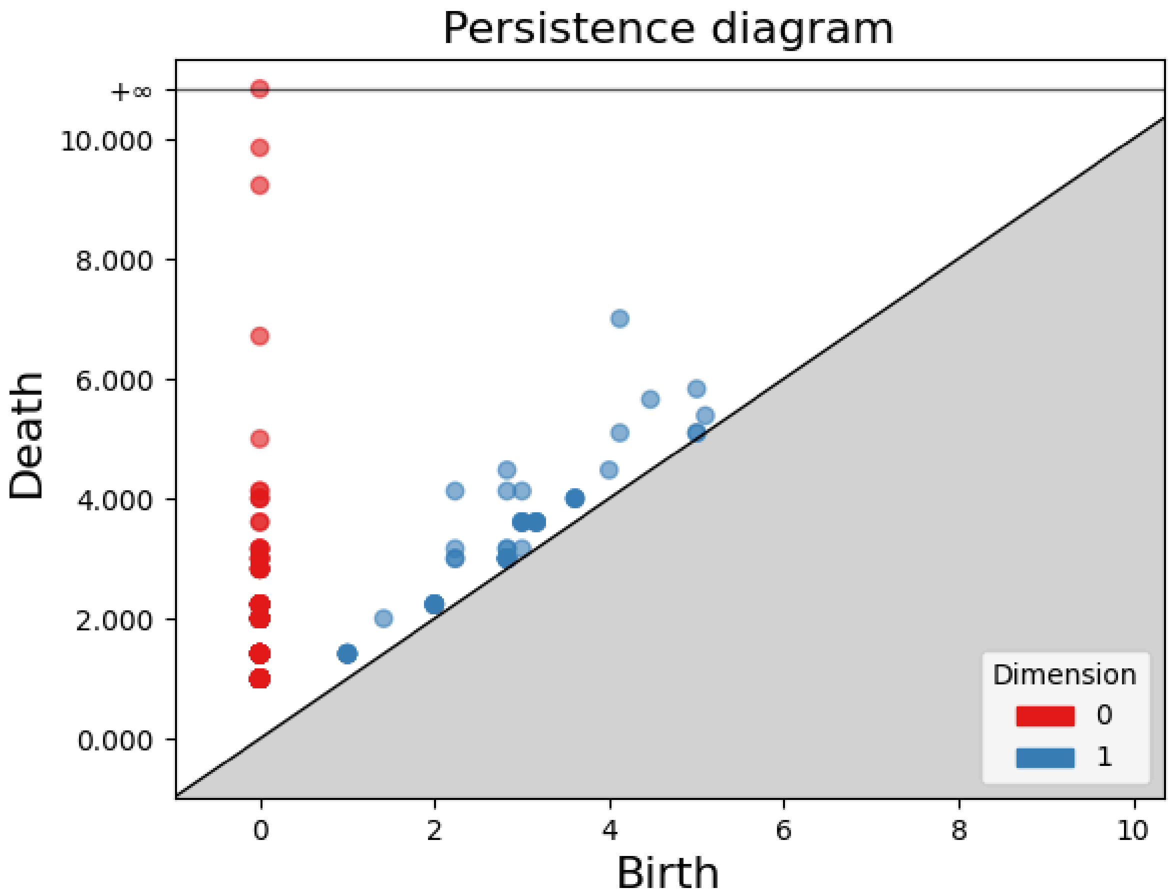


Persistence Diagram for Cloud A

In [19]:
```
import gudhi.representations

# Build the Rips complex
rips_complex = gd.RipsComplex(points=cloud_A, max_edge_length=20.0)
simplex_tree = rips_complex.create_simplex_tree(max_dimension=2)

# Compute persistence diagram
diag = simplex_tree.persistence()

# Filter out dimension 0 or 1 features
persistence_pairs = simplex_tree.persistence_intervals_in_dimension(1)

# Plot the persistence diagram
gd.plot_persistence_diagram(diag)
```

```
/usr/local/lib/python3.11/dist-packages/gudhi/persistence_graphical_tools.p
y:129: UserWarning: usetex mode requires TeX.
  warnings.warn("usetex mode requires TeX.")
```

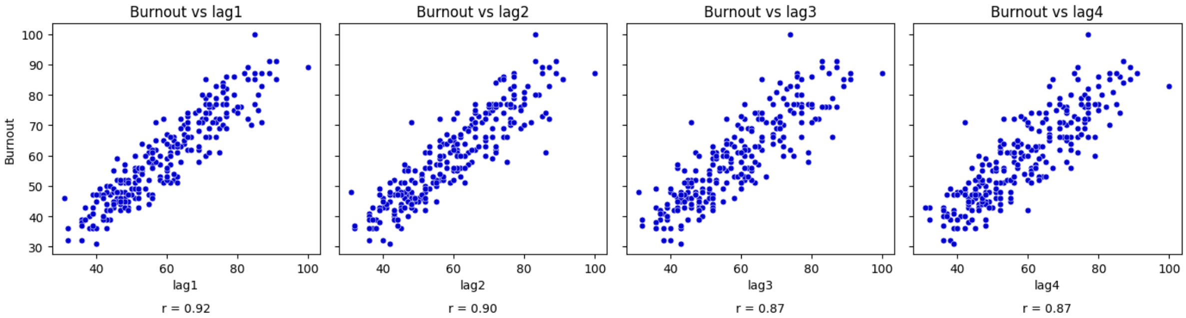
Out[19]:
```
<Axes: title={'center': 'Persistence diagram'}, xlabel='Birth', ylabel='Dea
th'>
```



Point Cloud B

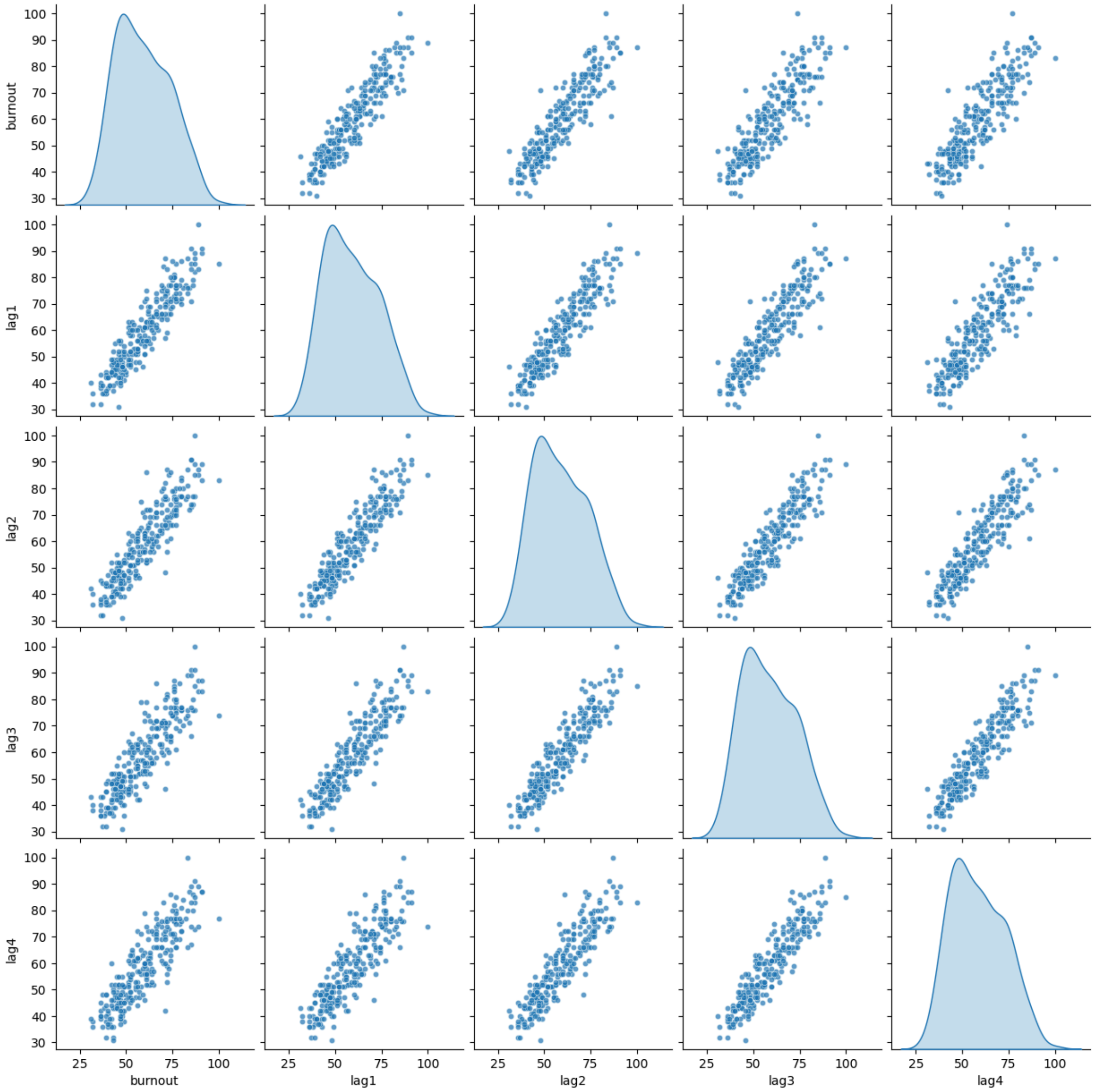
In [ ]:
```python
# Extract and prepare burnout series
burnout_series = data[['Date', 'burnout']].copy().dropna().reset_index(drop=

# Create lagged features (lags 1 to 4)
for lag in range(1, 5):
    burnout_series[f'lag{lag}'] = burnout_series['burnout'].shift(lag)

# Drop rows with NaNs introduced by lagging
burnout_lags = burnout_series.dropna().reset_index(drop=True)

# Define lag column names
lags = [f"lag{l}" for l in range(1, 5)]

# Create subplots
fig, axes = plt.subplots(1, 4, figsize=(14, 4), sharey=True)

# Plot each lag vs current burnout search interest
for i, lag in enumerate(lags):
    ax = axes[i]
    sb.scatterplot(
        data=burnout_lags,
        x=lag,
        y="burnout",
        s=25,
        color="mediumblue",
        ax=ax
    )
```

```
    # Axis labels
    ax.set_xlabel(f"{lag}")
    if i == 0:
        ax.set_ylabel("Burnout")
    else:
        ax.set_ylabel("")

    # Correlation annotation
    r = burnout_lags[lag].corr(burnout_lags["burnout"])
    ax.annotate(f"r = {r:.2f}", xy=(0.5, -0.25), xycoords="axes fraction",
                ha='center', fontsize=10)

    ax.set_title(f"Burnout vs {lag}")

# Layout & save
plt.tight_layout()
plt.savefig("burnout_lags_plot.png", dpi=300)
plt.show()
```



In [ ]:

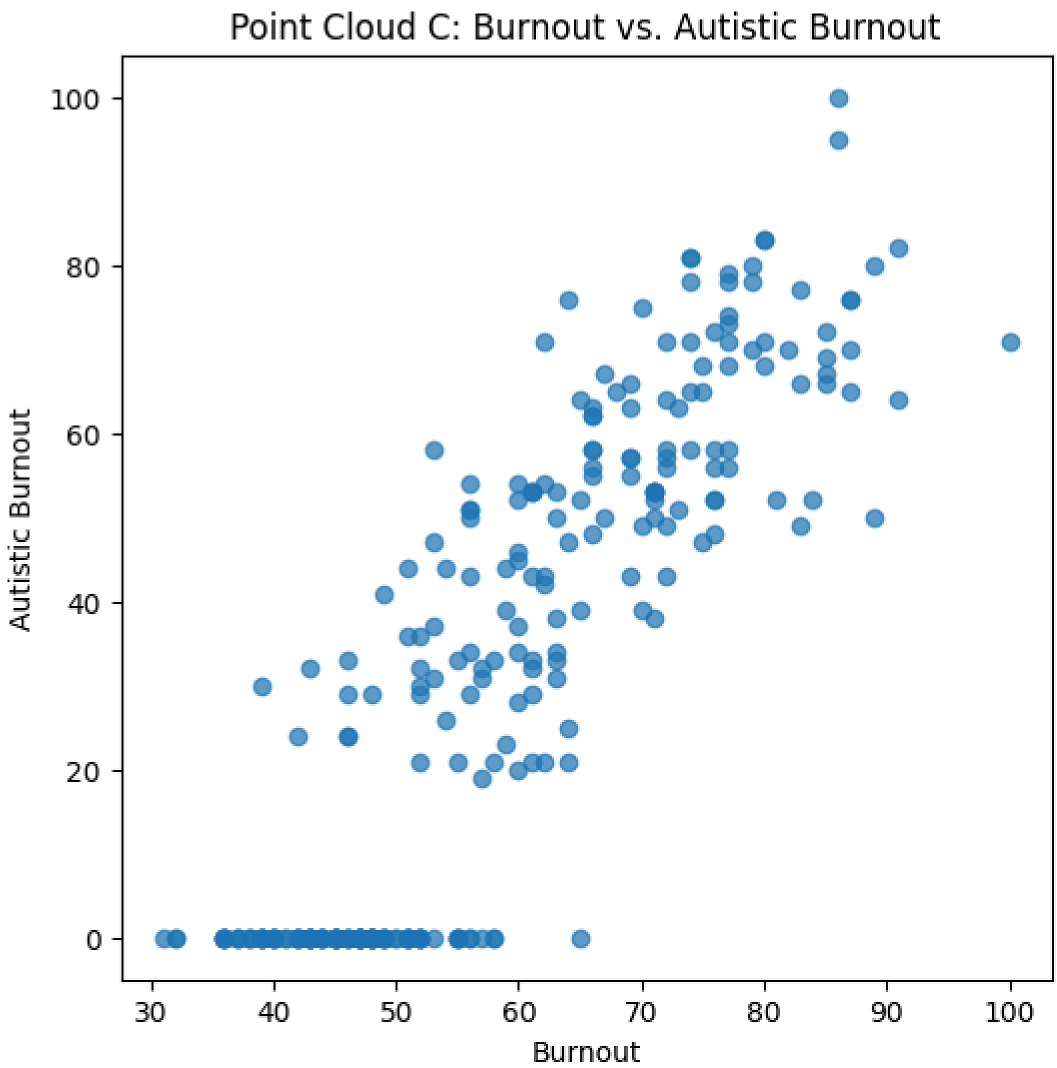
```
# Select only relevant columns: burnout and its lags
pairplot_data = burnout_lags[['burnout', 'lag1', 'lag2', 'lag3', 'lag4']]

# Create the pairplot
sns.pairplot(pairplot_data, diag_kind='kde', plot_kws={'s': 20, 'alpha': 0.7

# Save and show
plt.savefig("burnout_lags_pairplot.png", dpi=300)
plt.show()
```



Point Cloud C

```
In [ ]: # Drop missing values
point_cloud_C = data.dropna(subset=['burnout', 'autistic burnout'])

# Convert to numpy array
cloud_C = point_cloud_C[['burnout', 'autistic burnout']].to_numpy()

print("Shape of Cloud C:", cloud_C.shape)

import matplotlib.pyplot as plt

plt.figure(figsize=(6, 6))
plt.scatter(cloud_C[:, 0], cloud_C[:, 1], alpha=0.7)
plt.xlabel('Burnout')
plt.ylabel('Autistic Burnout')
plt.title('Point Cloud C: Burnout vs. Autistic Burnout')
plt.show()
```

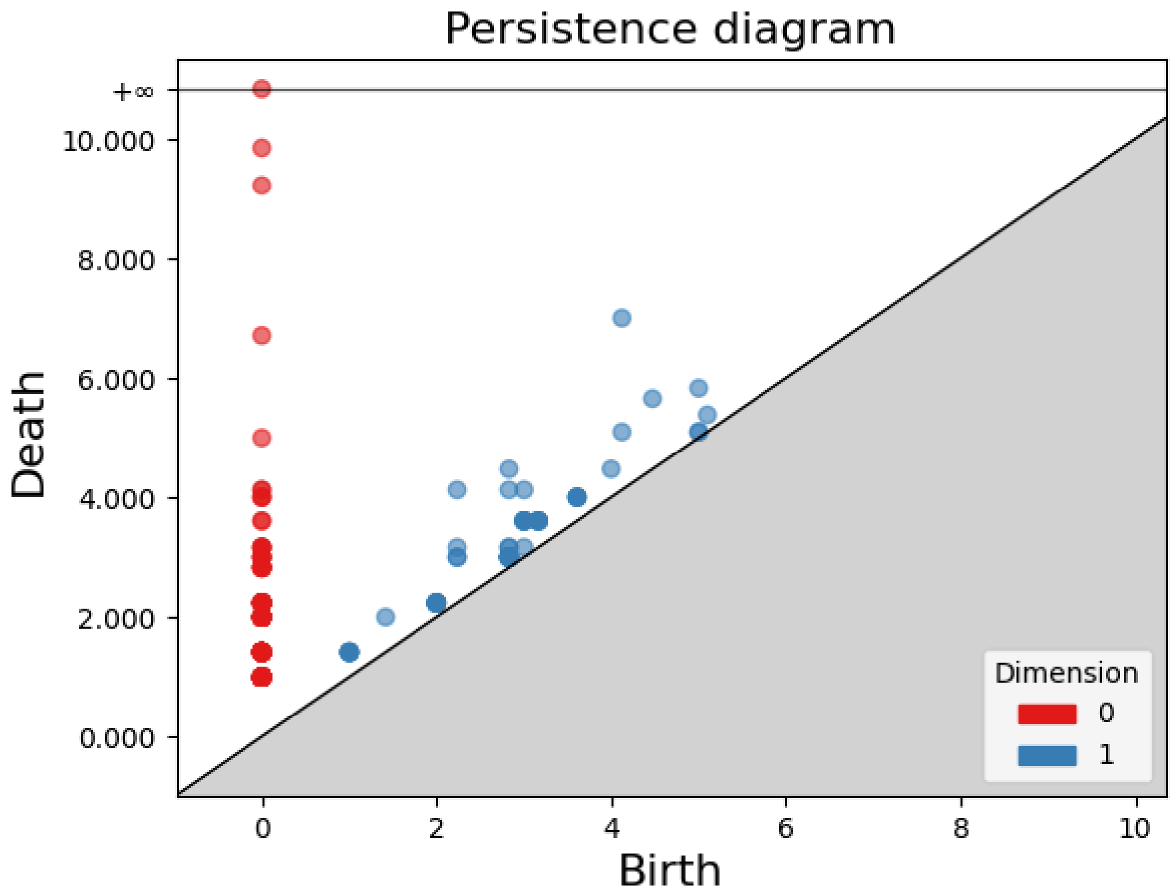
```
Shape of Cloud C: (261, 2)
```



Persistence Diagram for Point Cloud C

In [20]:
```
cloud_C = point_cloud[['burnout', 'burnout_lag1']].to_numpy()

# Step 1: Build the Rips complex
rips_complex = gd.RipsComplex(points=cloud_C, max_edge_length=20.0)
simplex_tree = rips_complex.create_simplex_tree(max_dimension=2)

# Step 2: Compute persistence diagram
diag = simplex_tree.persistence()

# Optional: Filter out dimension 0 or 1 features
persistence_pairs = simplex_tree.persistence_intervals_in_dimension(1)

# Step 3: Plot the persistence diagram
gd.plot_persistence_diagram(diag)
```

Out[20]:
```
<Axes: title={'center': 'Persistence diagram'}, xlabel='Birth', ylabel='Dea
th'>
```



In [ ]:
```
data.head()
```

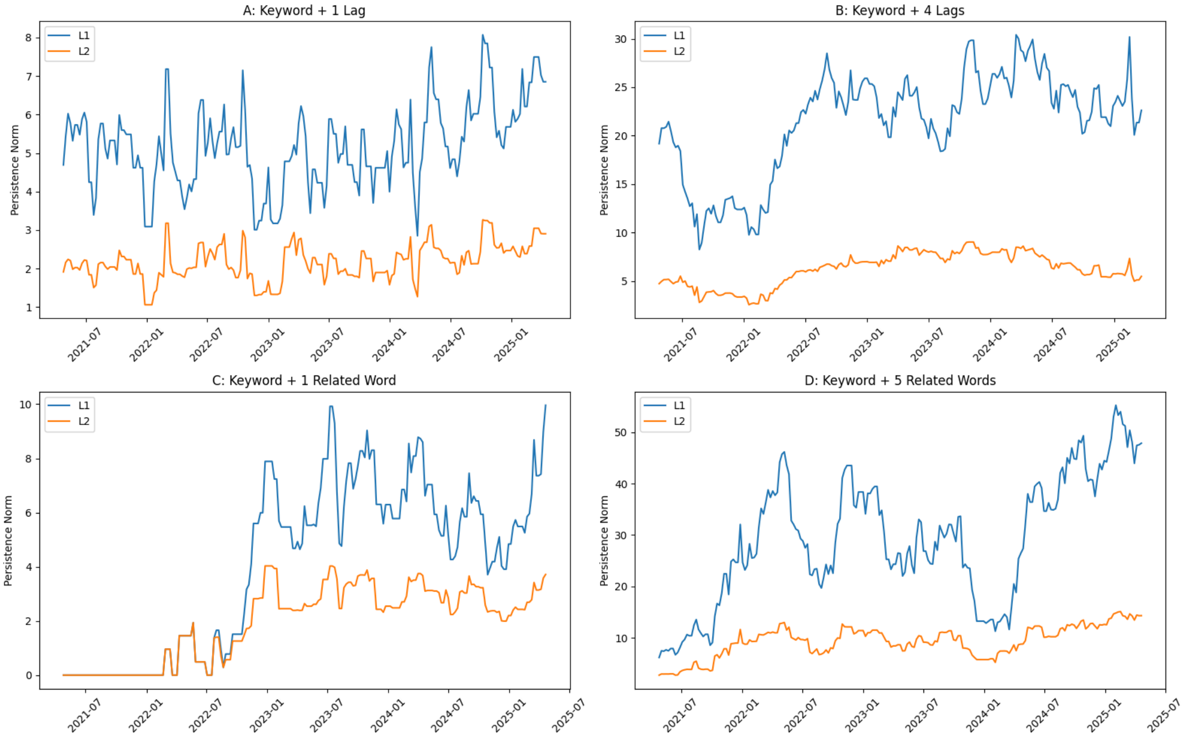
Out[ ]:

	Date	burnout	job stress	emotional exhaustion	depression	autistic burnout	work burnout	Month	burnout_lag1
0	2020-04-26	40	43	0	90	0	0	4	NaN
1	2020-05-03	38	28	0	90	0	0	5	40.0
2	2020-05-10	37	36	0	88	0	33	5	38.0
3	2020-05-17	38	35	0	97	0	0	5	37.0
4	2020-05-24	36	43	0	83	0	36	5	38.0

Time Series of Persistence Norms

In [8]:
```
def compute_persistence_norms_and_diagrams(embed2, window_length=52, max_di
    r, c = embed2.shape
    r2 = r - window_length
    norms = []
    diagrams_dim_1 = []

    for i in range(r2):
        window = embed2.iloc[i:i + window_length, :].to_numpy()
        rips = gd.RipsComplex(points=window, max_edge_length=radius)
        simplex_tree = rips.create_simplex_tree(max_dimension=max_dim)
```

```
        pers = simplex_tree.persistence()
        pers = [p for p in pers if np.isfinite(p[1][0]) and np.isfinite(p[1]

        # Save dimension 1 diagram
        diag_dim_1 = [p[1] for p in pers if p[0] == 1]
        diagrams_dim_1.append(diag_dim_1)

        # Compute persistence norms
        L10 = sum(p[1][1] - p[1][0] for p in pers if p[0] == 0)
        L20 = math.sqrt(sum((p[1][1] - p[1][0]) ** 2 for p in pers if p[0] =
        L11 = sum(p[1][1] - p[1][0] for p in pers if p[0] == 1)
        L21 = math.sqrt(sum((p[1][1] - p[1][0]) ** 2 for p in pers if p[0] =

        norms.append([i + window_length, L10, L11, L20, L21])

    df = pd.DataFrame(norms, columns=['Obs', 'L10', 'L11', 'L20', 'L21'])
    return df, diagrams_dim_1

def plot_norms_subplot(norms_df, merged_df, ax, title):
    dff = merged_df[['Date']].copy()
    dff['Obs'] = dff.index
    merged = dff.merge(norms_df, on='Obs')

    df_long = merged.melt(id_vars=['Date'], value_vars=['L11', 'L21'],
                          var_name='Type', value_name='Persistence Norm')
    df_long['Type'] = df_long['Type'].replace({'L11': 'L1', 'L21': 'L2'})

    sb.lineplot(x='Date', y='Persistence Norm', hue='Type', data=df_long, ax
    ax.set_title(title)
    ax.set_xlabel('')
    ax.set_ylabel('Persistence Norm')
    ax.tick_params(axis='x', rotation=45)
    ax.legend().set_title('')

# Create embeddings
embed_A = pd.concat([data['burnout'], data['burnout'].shift(1)], axis=1).dro
embed_B = pd.concat([data['burnout']] + [data['burnout'].shift(i) for i in r
embed_C = pd.concat([data['burnout'], data['autistic burnout']], axis=1).dro
related_cols = ['burnout', 'job stress', 'emotional exhaustion', 'depressio
embed_D = data[related_cols].dropna()

# Compute norms and save diagrams
norms_A, diagrams_A = compute_persistence_norms_and_diagrams(embed_A)
norms_B, diagrams_B = compute_persistence_norms_and_diagrams(embed_B)
norms_C, diagrams_C = compute_persistence_norms_and_diagrams(embed_C)
norms_D, diagrams_D = compute_persistence_norms_and_diagrams(embed_D)

# Plot
fig, axs = plt.subplots(2, 2, figsize=(16, 10))

plot_norms_subplot(norms_A, data.reset_index(drop=True), axs[0, 0], 'A: Keyw
plot_norms_subplot(norms_B, data.reset_index(drop=True), axs[0, 1], 'B: Keyw
plot_norms_subplot(norms_C, data.reset_index(drop=True), axs[1, 0], 'C: Keyw
plot_norms_subplot(norms_D, data.reset_index(drop=True), axs[1, 1], 'D: Keyw

plt.tight_layout()
plt.show()
```



Time Series of Wasserstein Distance

In [15]:
```
from gudhi.wasserstein import wasserstein_distance

# Compute persistence diagrams
def compute_persistence_diagrams(embed, window_length=52, max_dim=2, radius=
    diagrams = []
    for i in range(len(embed) - window_length):
        window = embed.iloc[i:i + window_length].to_numpy()
        rips = gd.RipsComplex(points=window, max_edge_length=radius)
        st = rips.create_simplex_tree(max_dimension=max_dim)
        pers = st.persistence()
        pers = [p for p in pers if np.isfinite(p[1][0]) and np.isfinite(p[1]
        diag_1 = [p[1] for p in pers if p[0] == 1]
        if not diag_1:
            diagrams.append(np.empty((0, 2)))
        else:
            diagrams.append(np.array(diag_1))
    return diagrams

# Compute Wasserstein time series
def compute_wasserstein_series(diagrams):
    distances = []
    for i in range(len(diagrams) - 1):
        d1 = diagrams[i]
        d2 = diagrams[i + 1]
        dist = wasserstein_distance(d1, d2, order=1., internal_p=2.)
        distances.append(dist)
    return distances

# Embeddings for A, B, C, D
embed_A = pd.concat([data['burnout'], data['burnout'].shift(1)], axis=1).dro
embed_B = pd.concat([data['burnout']] + [data['burnout'].shift(i) for i in r
embed_C = pd.concat([data['burnout'], data['autistic burnout']], axis=1).dro
embed_D = data[['burnout', 'job stress', 'emotional exhaustion', 'depression

# Diagrams
diagrams_A = compute_persistence_diagrams(embed_A)
diagrams_B = compute_persistence_diagrams(embed_B)
```

```
diagrams_C = compute_persistence_diagrams(embed_C)
diagrams_D = compute_persistence_diagrams(embed_D)

# Wasserstein distances
wd_A = compute_wasserstein_series(diagrams_A)
wd_B = compute_wasserstein_series(diagrams_B)
wd_C = compute_wasserstein_series(diagrams_C)
wd_D = compute_wasserstein_series(diagrams_D)

# Calculate the minimum length of the Wasserstein distance series
min_len = min(len(wd_A), len(wd_B), len(wd_C), len(wd_D))

# Adjust the starting index to account for lag and windowing
start_index = 53 + (len(wd_A) - min_len)

# Slice the date index to the minimum length
base_date_index = data['Date'][start_index : start_index + min_len].reset_in

# Truncate Wasserstein distance series to the minimum lengths
wd_A = wd_A[-(min_len):]
wd_B = wd_B[-(min_len):]
wd_C = wd_C[-(min_len):]
wd_D = wd_D[-(min_len):]

# Create DataFrame for plotting
wasserstein_df = pd.DataFrame({
    'Date': base_date_index,
    'A: 1 Lag': wd_A,
    'B: 4 Lags': wd_B,
    'C: 1 Related': wd_C,
    'D: 5 Related': wd_D
})

# Melt and Plot
df_long = wasserstein_df.melt(id_vars='Date', var_name='Embedding', value_na
plt.figure(figsize=(12, 6))
sb.lineplot(data=df_long, x='Date', y='Wasserstein Distance', hue='Embedding
plt.xticks(rotation=45)
plt.title('Wasserstein Distances Between Persistence Diagrams (Weekly Increm
plt.tight_layout()
plt.show()
```